Assignment 2: Launching, Load balancing & Scaling on AWS EC2

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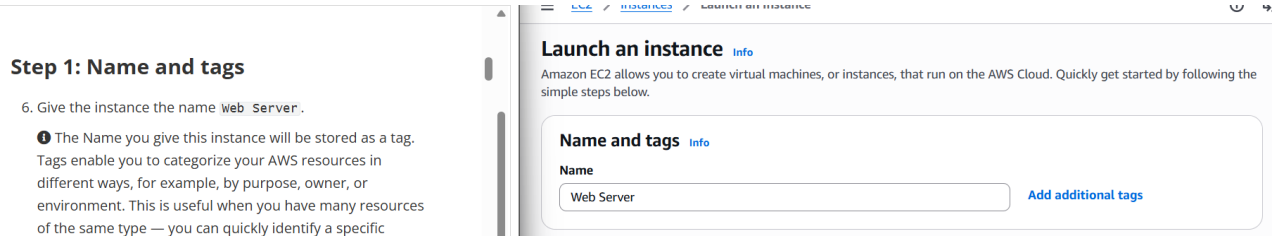
Overview

This assignment provides hands-on experience with core AWS services and modern cloud computing patterns. You will start by launching and managing EC2 instances using AWS Console. Building on this foundation, you will configure load balancing and auto-scaling groups to understand how AWS handles fault tolerance and dynamic scaling under traffic. Finally, you will reflect on the trade-offs between AWS services to make informed architectural decisions for real-world cloud applications.

Task 1 (Access Module 6 Lab - 3 Introduction to Amazon EC2 from AWS academy account) – Launching an EC2 Instance using AWS console

- Name and tag:

The Name you give this instance will be stored as a tag. Tags enable you to categorize your AWS resources in different ways, for example, by purpose, owner, or environment



- Application and OS Images (Amazon Machine Image)

In the list of available Quick Start AMIs, keep the default Amazon Linux AMI selected

An Amazon Machine Image (AMI) provides the information required to launch an instance, which is a virtual server in the cloud

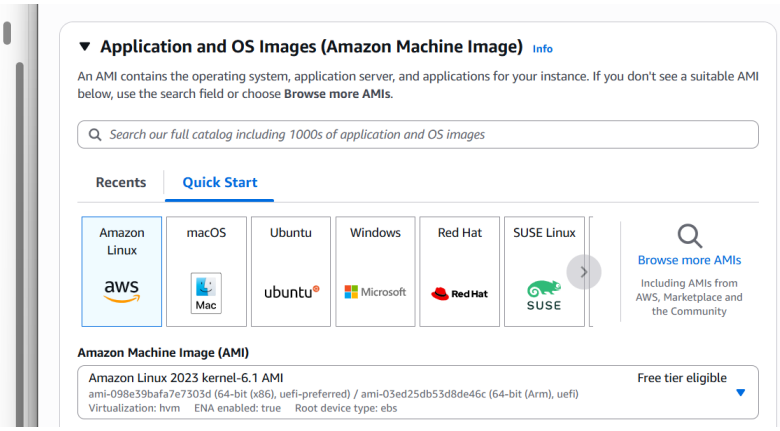
Step 2: Application and OS Images (Amazon Machine Image)

7. In the list of available *Quick Start* AMIs, keep the default **Amazon Linux** AMI selected.

8. Also keep the default **Amazon Linux 2023 AMI** selected.

   An **Amazon Machine Image (AMI)** provides the information required to launch an instance, which is a virtual server in the cloud. An AMI includes:

   - A template for the root volume for the instance (for example, an operating system or an application server with applications)
   - Launch permissions that control which AWS accounts can use the AMI to launch instances
   - A block device mapping that specifies the volumes to



- Instance type

In the Instance type panel, keep the default t2.micro selected

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases

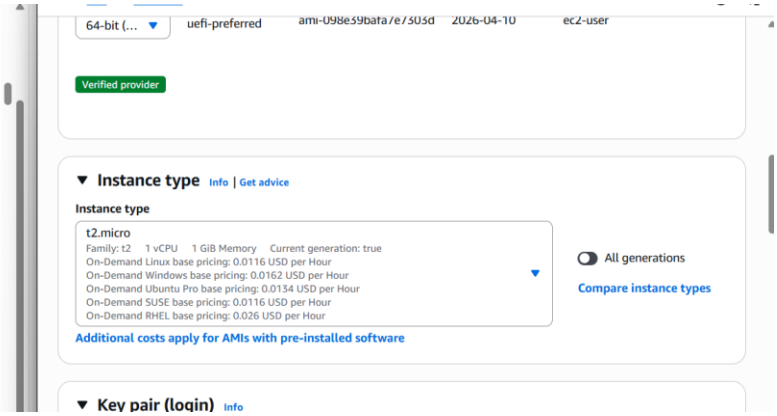
Step 3: Instance type

9. In the *Instance type* panel, keep the default **t2.micro** selected.

   Amazon EC2 provides a wide selection of *instance types* optimized to fit different use cases. Instance types comprise varying combinations of CPU, memory, storage, and networking capacity and give you the flexibility to choose the appropriate mix of resources for your applications. Each instance type includes one or more *instance sizes*, allowing you to scale your resources to the requirements of your target workload.

   The t2.micro instance type has 1 virtual CPU and 1 GiB of memory.

   Note: You may be restricted from using other instance types in this lab.



- Key pair

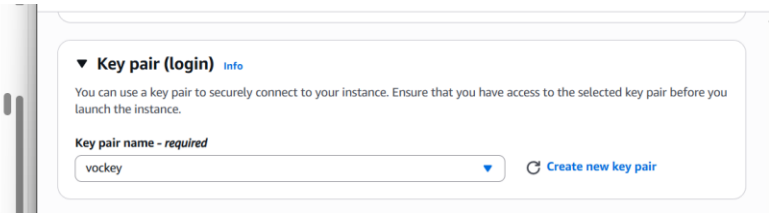
Amazon EC2 uses public–key cryptography to encrypt and decrypt login information

Amazon EC2 uses public–key cryptography to encrypt and decrypt login information

Step 4: Key pair (login)

10. For **Key pair name - *required***, choose **vockey**.

    Amazon EC2 uses public-key cryptography to encrypt and decrypt login information. To ensure you will be able to log in to the guest OS of the instance you create, you identify an existing key pair or create a new key pair when launching the instance. Amazon EC2 then installs the key on



- Network setting

The Lab VPC was created using an AWS CloudFormation template during the setup process of your lab. This VPC includes two public subnets in two different Availability Zones

Step 5: Network settings

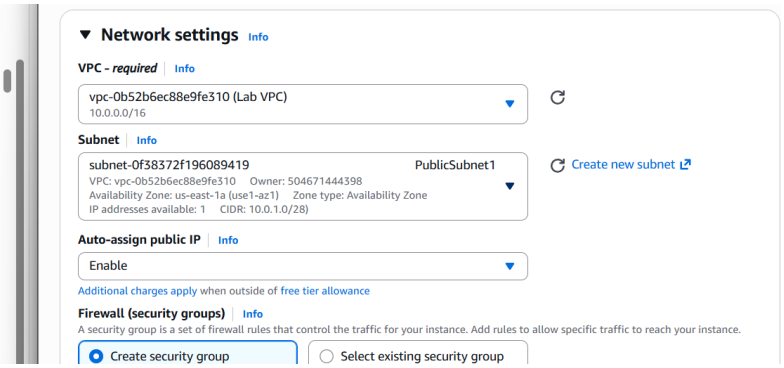
11. Next to Network settings, choose **Edit**.

12. For **VPC**, select **Lab VPC**.

    The Lab VPC was created using an AWS CloudFormation template during the setup process of your lab. This VPC includes two public subnets in two different Availability Zones.

    Note: Keep the default subnet **PublicSubnet1**. This is the subnet in which the instance will run. Notice also that by default, the instance will be assigned a public IP address.

13. Under **Firewall (security groups)**, choose **Create**



- Firewall setting

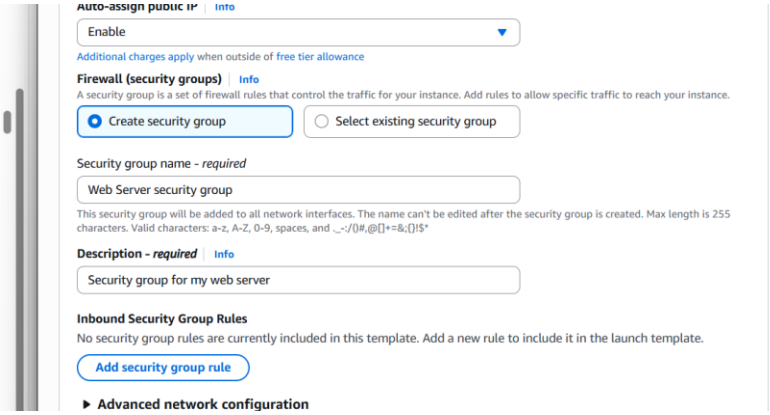
Security group name: Web Server security group

Description: Security group for my web server

Under Inbound security group rules, notice that one rule exists. Remove this rule

13. Under **Firewall (security groups)**, choose **Create security group** and configure:
    - **Security group name:** `Web Server security group`
    - **Description:** `Security group for my web server`

      A *security group* acts as a virtual firewall that controls the traffic for one or more instances. When you launch an instance, you associate one or more security groups with the instance. You add *rules* to each security group that allow traffic to or from its associated instances. You can modify the rules for a security group at any time; the new rules are automatically applied to all instances that are associated with the security group.
    - Under **Inbound security group rules**, notice that one rule exists. **Remove** this rule.



-Configure storage

Description: Security group for my web server

launch the Amazon EC2 instance using a default 8 GiB disk volume. This will be your root volume

Step 6: Configure storage

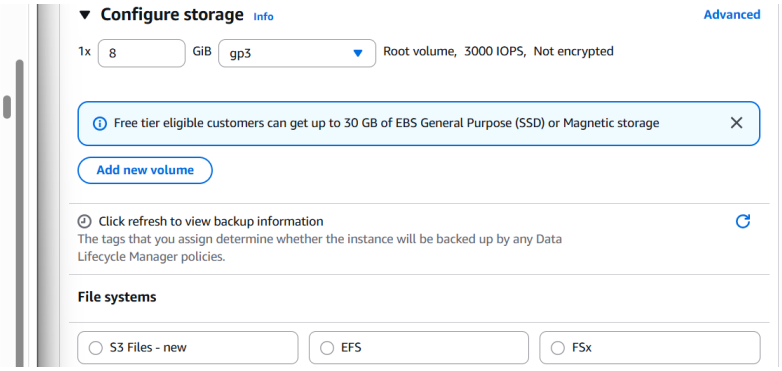
14. In the *Configure storage* section, keep the default settings.

    Amazon EC2 stores data on a network-attached virtual disk called *Elastic Block Store*.

    You will launch the Amazon EC2 instance using a default 8 GiB disk volume. This will be your root volume (also known as a 'boot' volume).

Step 7: Advanced details

15. Expand **Advanced details**.



- Advanced Detail

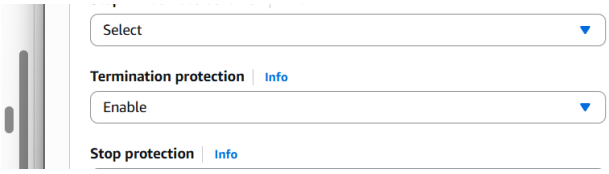
Expand -> Advanced details

For Termination protection, select Enable

For Termination protection, select Enable

16. For **Termination protection**, select **Enable**.

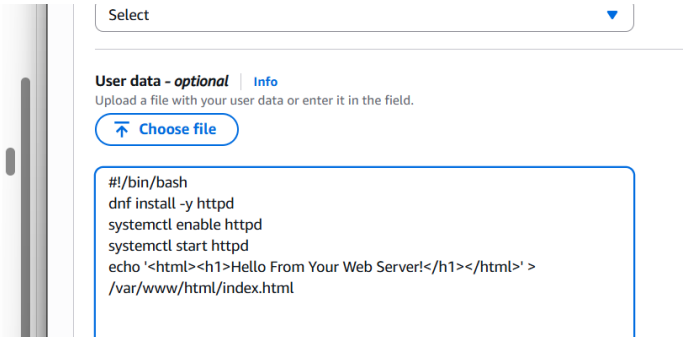
    When an Amazon EC2 instance is no longer required, it can be *terminated*, which means that the instance is deleted and its resources are released. A terminated instance cannot be accessed again and the data that was on it cannot be recovered. If you want to prevent the instance



17. Scroll to the bottom of the page and then copy and paste the code shown below into the **User data** box:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1>
</html>' > /var/www/html/index.html
```

    When you launch an instance, you can pass *user data* to the instance that can be used to perform automated



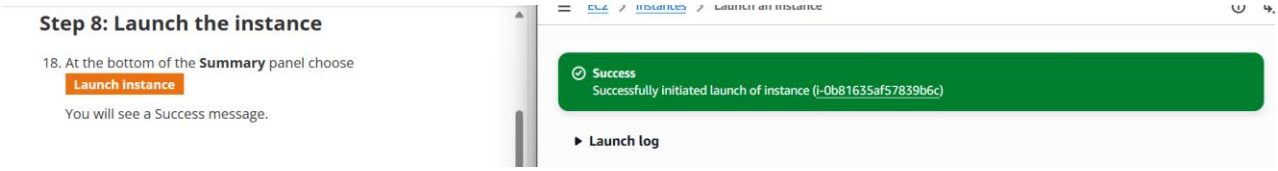
- Launch the instance

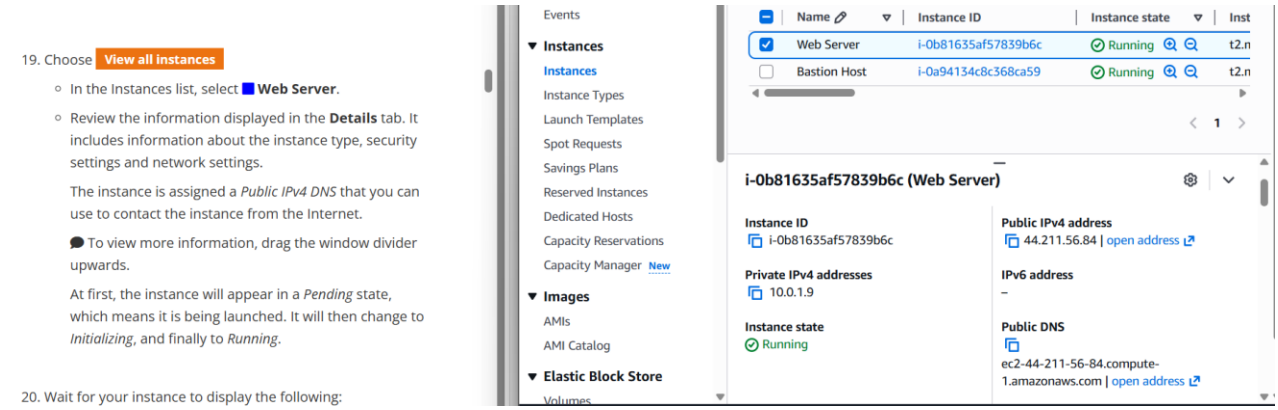
At the bottom of the **Summary** panel choose **Launch instance**

Choose **View all instances**

In the Instances list, select Web Server.

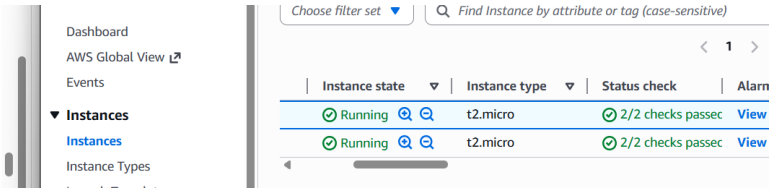
Review the information displayed in the Details tab





At first, the instance will appear in a *Pending* state, which means it is being launched. It will then change to *Initializing*, and finally to *Running*.

20. Wait for your instance to display the following:
    - **Instance State:** *Running*
    - **Status Checks:** *2/2 checks passed*



Task 2 (Access Module 6 Lab - 3 Introduction to Amazon EC2 from AWS academy account) – Access AWS EC2 instance through console for monitoring, managing activities.

-Monitor Your Instance

- Choose the status check

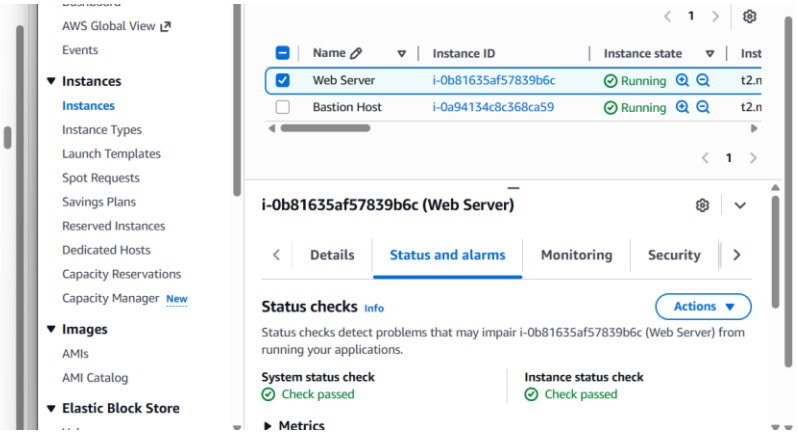
Task 2: Monitor Your Instance

Monitoring is an important part of maintaining the reliability, availability, and performance of your Amazon Elastic Compute Cloud (Amazon EC2) instances and your AWS solutions.

21. Choose the **Status checks** tab.

    With instance status monitoring, you can quickly determine whether Amazon EC2 has detected any problems that might prevent your instances from running applications. Amazon EC2 performs automated checks on every running EC2 instance to identify hardware and software issues.

    Notice that both the **System reachability** and **Instance reachability** checks have passed.



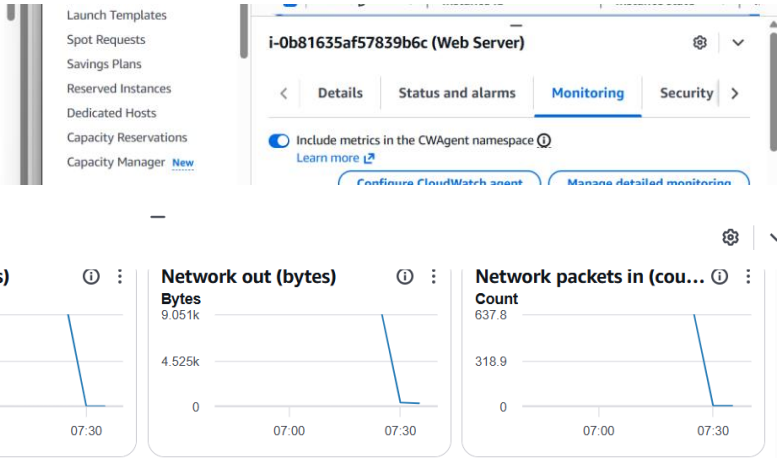
- Choose the monitoring tab

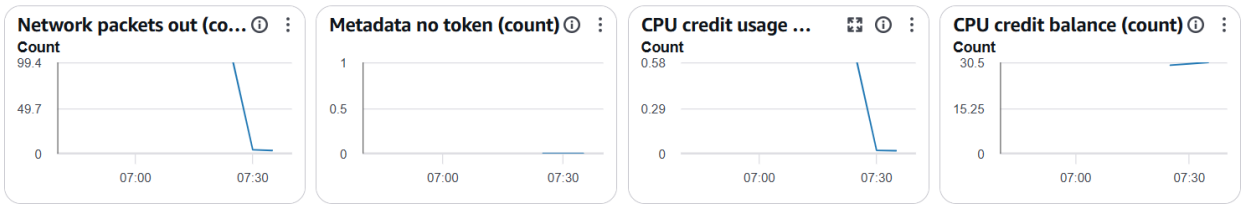
displays Amazon CloudWatch metrics for your instance

22. Choose the **Monitoring** tab.

    This tab displays Amazon CloudWatch metrics for your instance. Currently, there are not many metrics to display because the instance was recently launched.

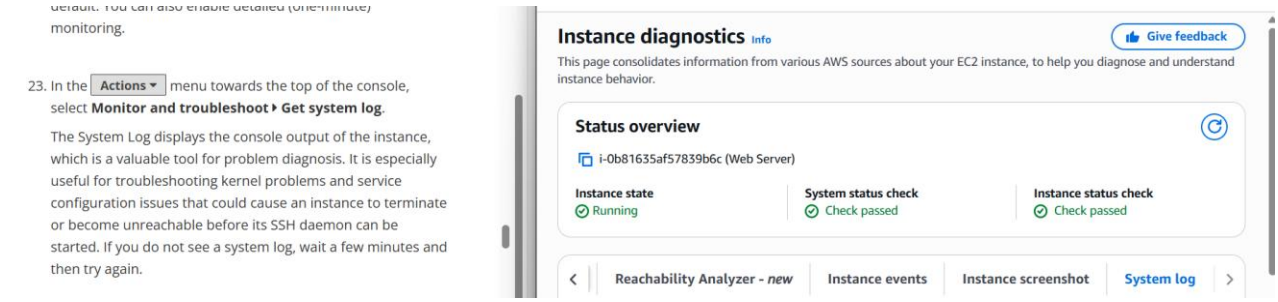
    You can choose the three dots icon in any graph and select **Enlarge** to see an expanded view of the chosen metric.





- Get system log

In the Actions menu towards the top of the console, select Monitor and troubleshoot -> Get system log



monitoring.

23. In the Actions menu towards the top of the console, select **Monitor and troubleshoot** ▸ **Get system log**.

    The System Log displays the console output of the instance, which is a valuable tool for problem diagnosis. It is especially useful for troubleshooting kernel problems and service configuration issues that could cause an instance to terminate or become unreachable before its SSH daemon can be started. If you do not see a system log, wait a few minutes and then try again.

Instance diagnostics Info

Give feedback

This page consolidates information from various AWS sources about your EC2 instance, to help you diagnose and understand instance behavior.

Status overview

i-0b81635af57839b6c (Web Server)

Instance state	System status check	Instance status check
Running	Check passed	Check passed

Reachability Analyzer - *new* | Instance events | Instance screenshot | System log



then try again.

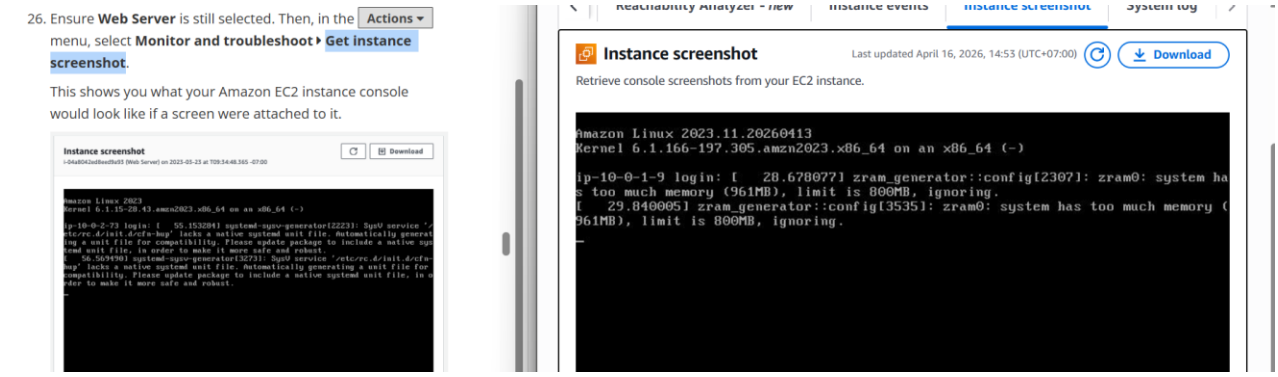
24. Scroll through the output and note that the HTTP package was installed from the **user data** that you added when you created the instance.

25. Choose **Cancel**.

EC2 > Instances > i-0b81635af57839b6c > Instance diagnostics

```
[   29.500763] cloud-init[2185]:   mod_http2-2.0.27-1.amzn2023.0.3.x86_64
[   29.500784] cloud-init[2185]:   mod_lua-2.4.66-1.amzn2023.0.1.x86_64
[   29.500804] cloud-init[2185]: Complete!
[   29.622466] cloud-init[2185]: Created symlink /etc/systemd/system/multi-user.target.wants/h
[   29.840005] zram_generator::config[3535]: zram0: system has too much memory (961MB), limit
ci-info: ++++++++++++++++++++++++++++++Authorized keys from /home/ec2-user/.ssh/authorized_keys f
ci-info: +---------+-------------------------------------------------------------------------+
ci-info: | Keytype |                                                         Fingerprint (sha256)
ci-info: +---------+-------------------------------------------------------------------------+
ci-info: | ssh-rsa | 38:41:e4:e0:20:ad:06:58:59:2e:01:54:c8:49:29:31:49:94:e2:97:db:35:9b:3a:8
ci-info: +---------+-------------------------------------------------------------------------+
<14>Apr 16 07:25:09 cloud-init: #############################################################
<14>Apr 16 07:25:09 cloud-init: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Apr 16 07:25:09 cloud-init: 256 SHA256:f9QQriyDGo4HuksBOv+hETNrA+rGS/WtH565ljjEwVk root@ip
<14>Apr 16 07:25:09 cloud-init: 256 SHA256:4LEZXV3V2QNOJDGbXlJ9jwrFcw7WrvlmaW5kYGZqnLw root@ip
<14>Apr 16 07:25:09 cloud-init: -----END SSH HOST KEY FINGERPRINTS-----
<14>Apr 16 07:25:09 cloud-init: #############################################################
-----BEGIN SSH HOST KEY KEYS-----
```

-Get instance screenshot



26. Ensure **Web Server** is still selected. Then, in the Actions menu, select **Monitor and troubleshoot** ▸ **Get instance screenshot**.

    This shows you what your Amazon EC2 instance console would look like if a screen were attached to it.

Instance screenshot

Last updated April 16, 2026, 14:53 (UTC+07:00)

Download

Retrieve console screenshots from your EC2 instance.

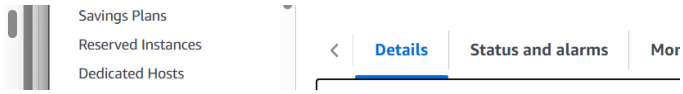
```
Amazon Linux 2023.11.20260413
Kernel 6.1.166-197.305.amzn2023.x86_64 on an x86_64 (-)

ip-10-0-1-9 login: [   28.678077] zram_generator::config[2307]: zram0: system ha
s too much memory (961MB), limit is 800MB, ignoring.
[   29.840005] zram_generator::config[3535]: zram0: system has too much memory (
961MB), limit is 800MB, ignoring.
```

- Update Your Security Group and Access the Web Server

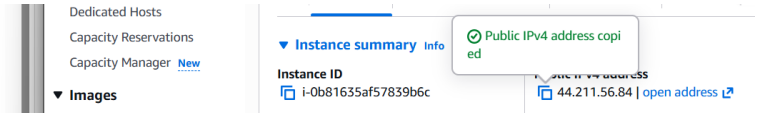
- Detail tab

28. Ensure **Web Server** is still selected. Choose the **Details** tab.

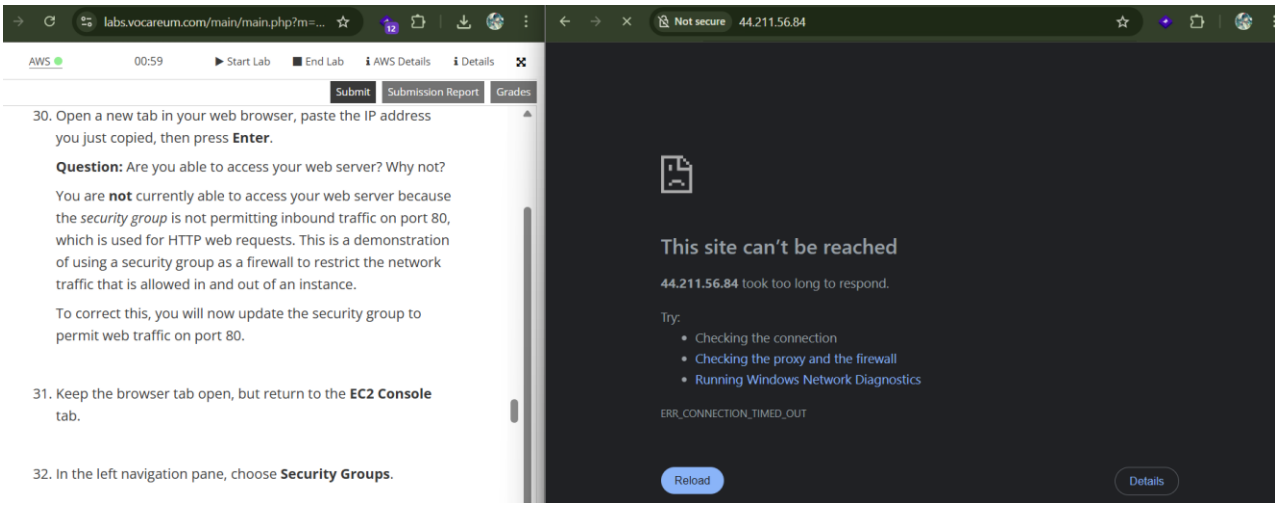


- Copy ipv4

29. Copy the **Public IPv4 address** of your instance to your clipboard.



- Paste ipv4 url

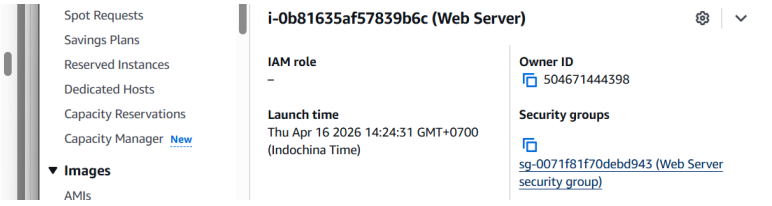


- Web server security service

31. Keep the browser tab open, but return to the **EC2 Console** tab.

32. In the left navigation pane, choose **Security Groups**.

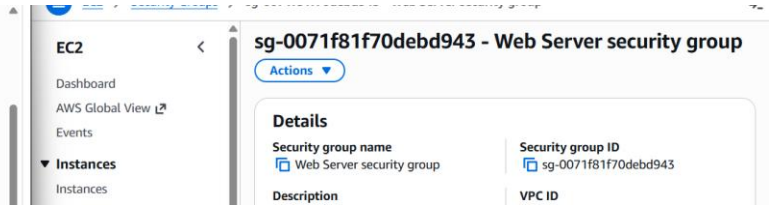
33. Select **Web Server security group**.



31. Keep the browser tab open, but return to the **EC2 Console** tab.

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33. Select **Web Server security group**.



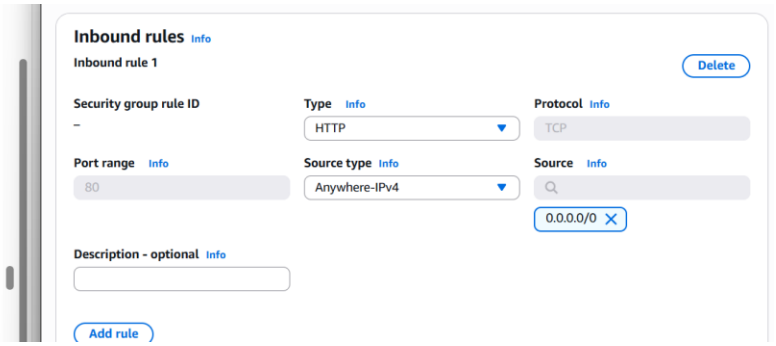
- Select inbound rule

33. Select **Web Server security group**.

34. Choose the **Inbound rules** tab.

    The security group currently has no inbound rules.

35. Choose **Edit inbound rules**, select **Add rule** and then configure:
    - **Type:** *HTTP*
    - **Source:** *Anywhere-IPv4*
    - Choose **Save rules**



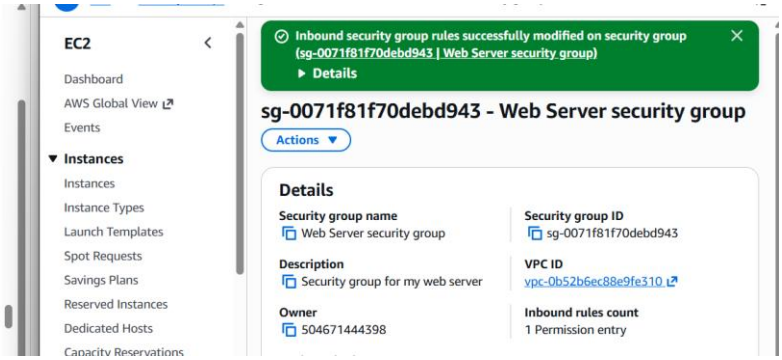
32. In the left navigation pane, choose **Security Groups**.

33. Select **Web Server security group**.

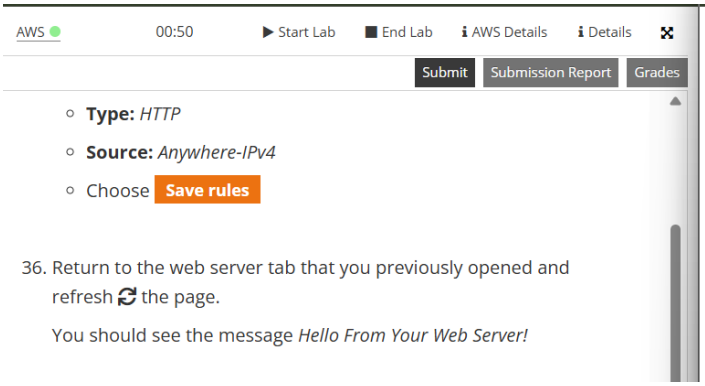
34. Choose the **Inbound rules** tab.

    The security group currently has no inbound rules.

35. Choose **Edit inbound rules**, select **Add rule** and then configure:
    - **Type:** *HTTP*
    - **Source:** *Anywhere-IPv4*
    - Choose **Save rules**



- Refresh page





- Resize Your Instance: Instance Type and EBS Volume

- Stop instance

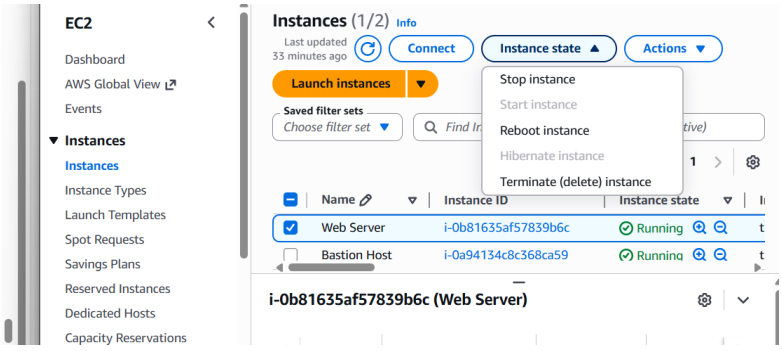
Stop Your Instance

Before you can resize an instance, you must *stop* it.

When you stop an instance, it is shut down. There is no runtime charge for a stopped EC2 instance, but the storage charge for attached Amazon EBS volumes remains.

37. On the **EC2 Management Console**, in the left navigation pane, choose **Instances** and then select the **Web Server** instance.

38. In the **Instance state** menu, select **Stop instance**.



When you stop an instance, it is shut down. There is no runtime charge for a stopped EC2 instance, but the storage charge for attached Amazon EBS volumes remains.

37. On the **EC2 Management Console**, in the left navigation pane, choose **Instances** and then select the **Web Server** instance.

38. In the Instance state menu, select **Stop instance**.

39. Choose Stop

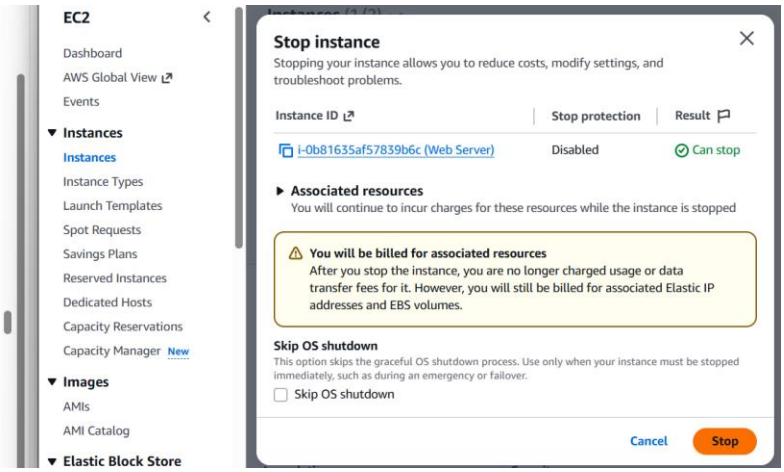
    Your instance will perform a normal shutdown and then will stop running.

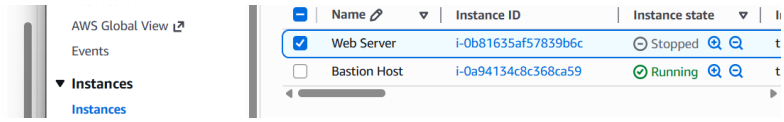
40. Wait for the **Instance state** to display: *Stopped*.

Change The Instance Type and enable

stop running.

40. Wait for the **Instance state** to display: *Stopped*.





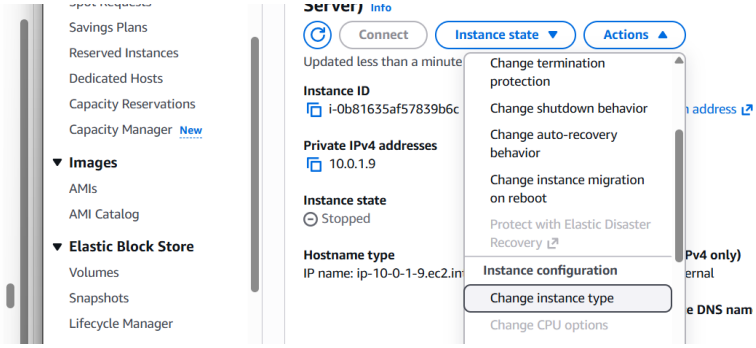
-Change The Instance Type and enable stop protection

- Change instance type

Change The Instance Type and enable stop protection

41. Select the Web Server instance, then in the Actions menu, select **Instance settings** ▸ **Change instance type**, then configure:
    - **Instance Type:** *t2.small*
    - Choose Apply

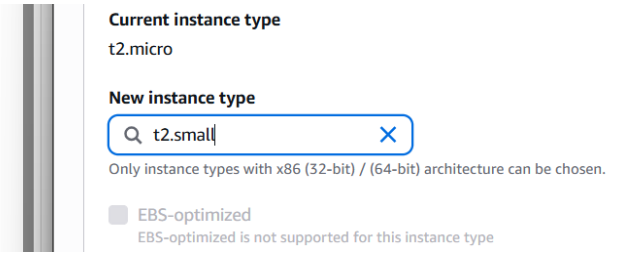
      When the instance is started again it will run as a *t2.small*, which has twice as much memory as a *t2.micro* instance. **NOTE**: You may be restricted from using other instance types in this lab.

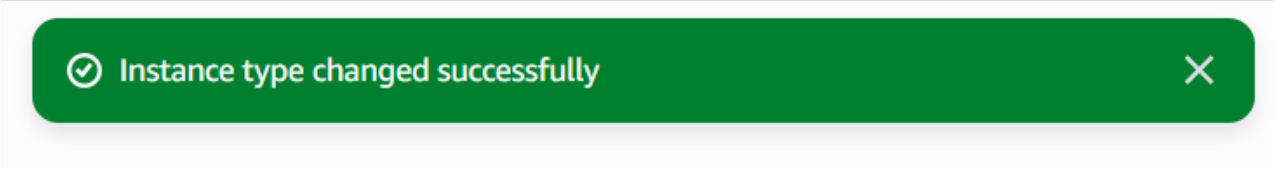


configure:

- **Instance Type:** *t2.small*
- Choose Apply

  When the instance is started again it will run as a *t2.small*, which has twice as much memory as a *t2.micro* instance. **NOTE**: You may be restricted from using other instance types in this lab.

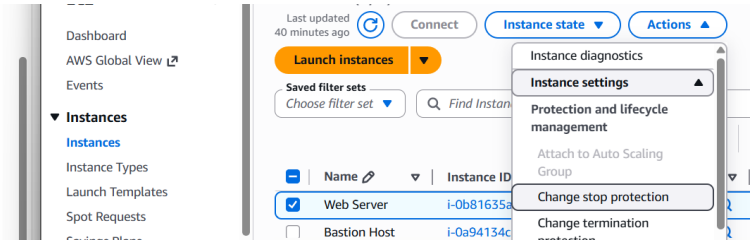


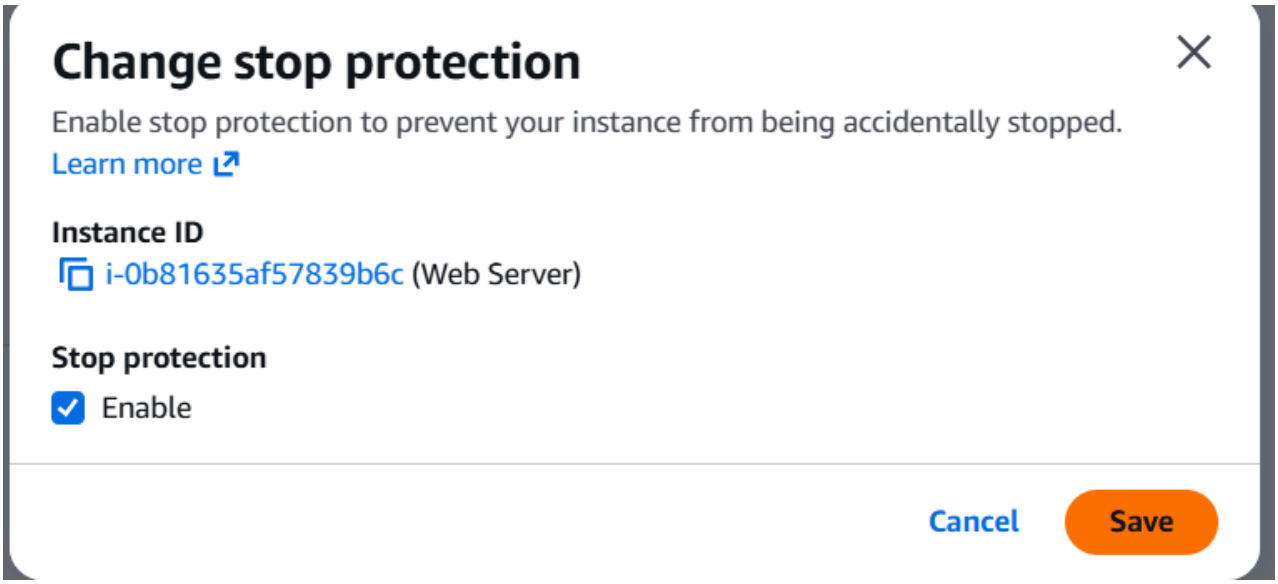


types in this lab.

42. Select the Web Server instance, then in the **Actions** menu, select **Instance settings** ▸ **Change stop protection**. Select **Enable** and then **Save** the change.

    When you stop an instance, the instance shuts down. When you later start the instance, it is typically migrated to a new underlying host computer and assigned a new *public* IPv4 address. An instance retains its assigned *private* IPv4 address. When you stop an instance, it is not deleted. Any EBS volumes





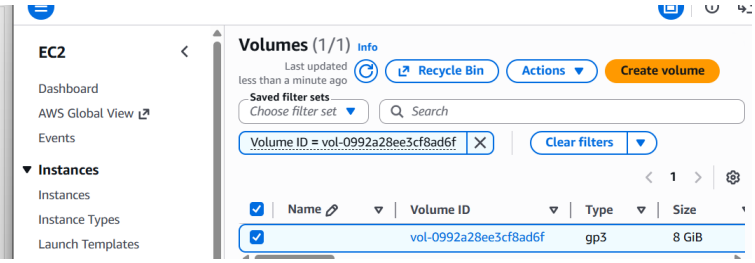
-Resize the EBS Volume

Resize the EBS Volume

43. With the Web Server instance still selected, choose the **Storage** tab, select the name of the Volume ID, then select the checkbox next to the volume that displays.

44. In the **Actions** menu, select **Modify volume**.

    The disk volume currently has a size of 8 GiB. You will now increase the size of this disk.



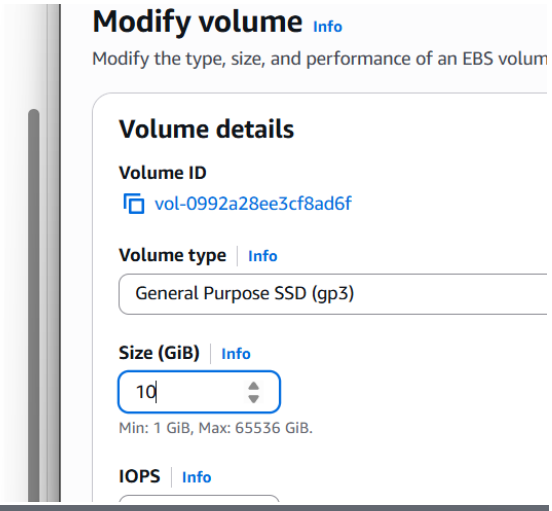
43. With the Web Server instance still selected, choose the **Storage** tab, select the name of the Volume ID, then select the checkbox next to the volume that displays.

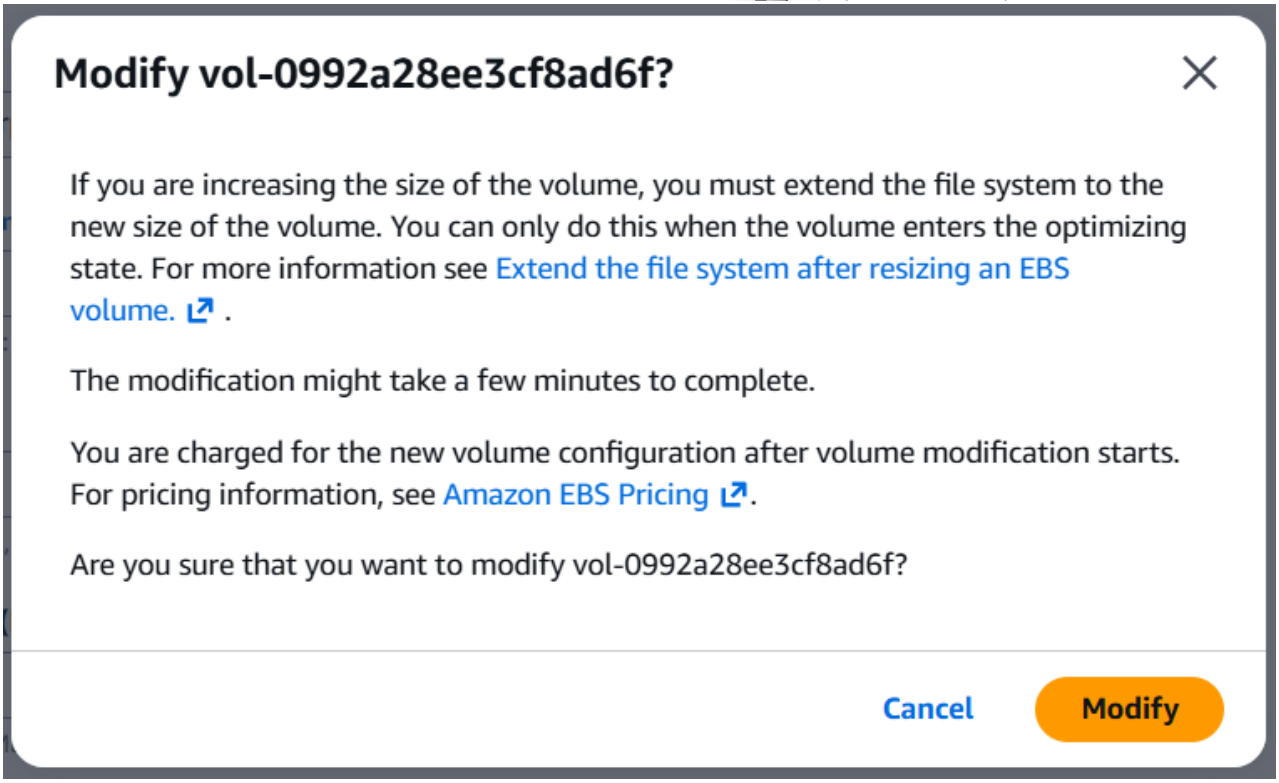
44. In the **Actions** menu, select **Modify volume**.

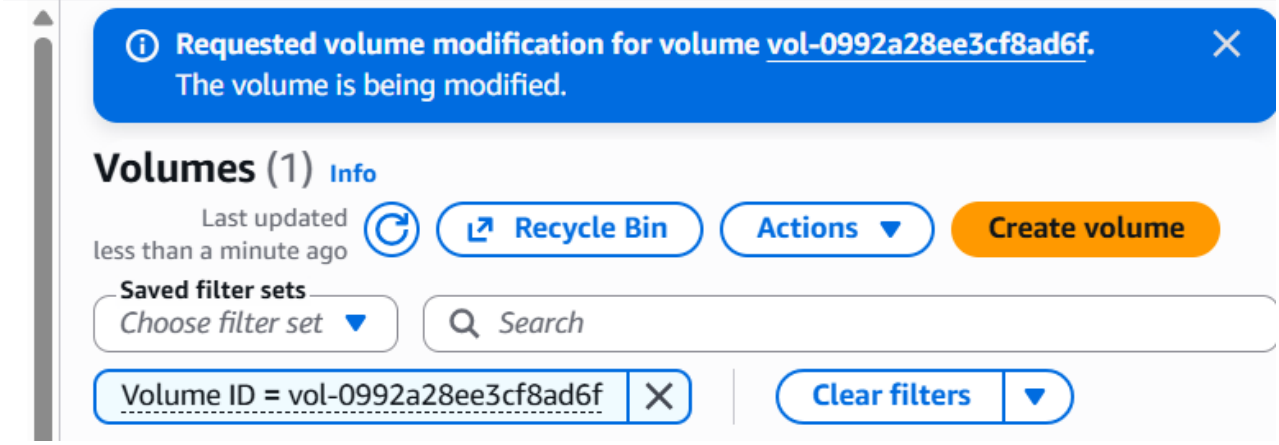
    The disk volume currently has a size of 8 GiB. You will now increase the size of this disk.

45. Change the size to: `10` **NOTE**: You may be restricted from creating Amazon EBS volumes larger than 10 GB in this lab.

46. Choose **Modify**







-Start the Resized Instance

Start the Resized Instance

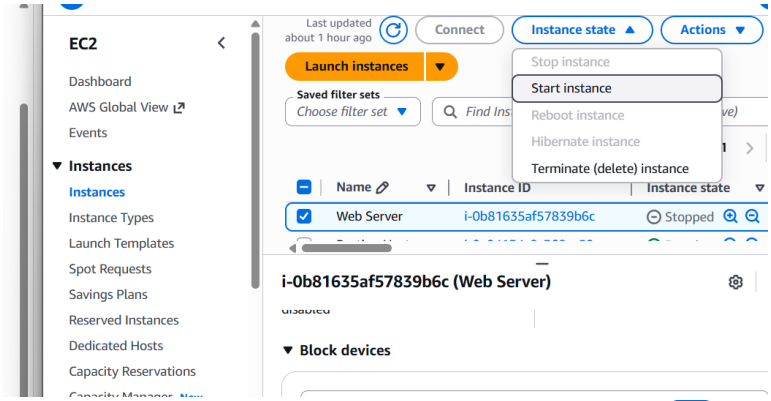
You will now start the instance again, which will now have more memory and more disk space.

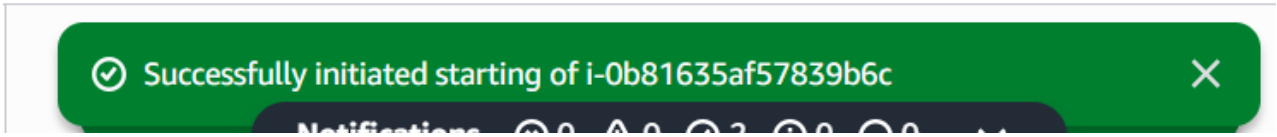
48. In left navigation pane, choose **Instances**.

49. Select the **Web Server** instance.

50. In the **Instance state** menu, select **Start instance**.

    Congratulations! You have successfully resized your Amazon EC2 Instance. In this task you changed your instance type from *t2.micro* to *t2.small*. You also modified your root disk volume from 8 GiB to 10 GiB.





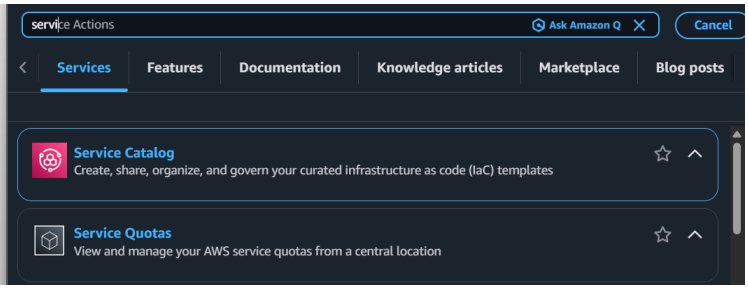
- Explore EC2 Limits

- Service quota

Task 5: Explore EC2 Limits

Amazon EC2 provides different resources that you can use. These resources include images, instances, volumes, and snapshots. When you create an AWS account, there are default limits on these resources on a per-region basis.

51. In the AWS Management Console, in the search box next to **Services**, search for and choose `Service Quotas`

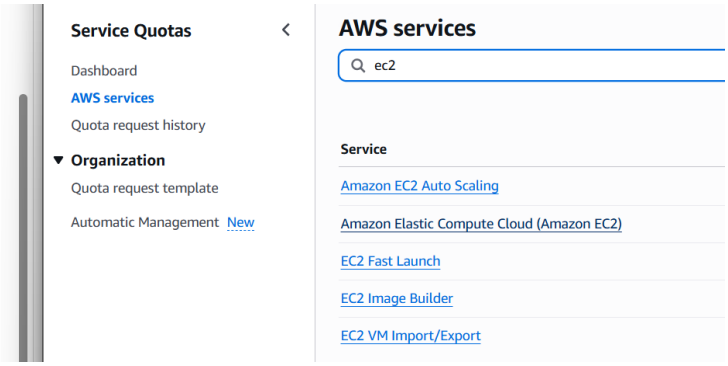


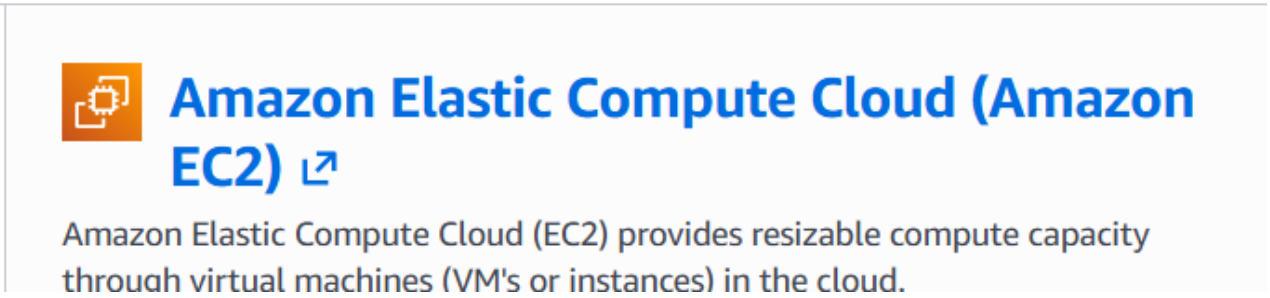
- Amazon Elastic Compute Cloud (Amazon EC2)

Amazon EC2 provides different resources that you can use. These resources include images, instances, volumes, and snapshots. When you create an AWS account, there are default limits on these resources on a per-region basis.

51. In the AWS Management Console, in the search box next to **Services**, search for and choose `Service Quotas`

52. Choose **AWS services** from the navigation menu and then in the AWS services *Find services* search bar, search for `ec2` and choose **Amazon Elastic Compute Cloud (Amazon EC2)**.

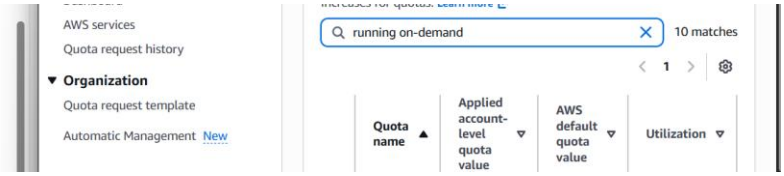




- find `running on-demand`

53. In the *Find quotas* search bar, search for `running on-demand`, but do not make a selection. Instead, observe the filtered list of service quotas that match the criteria.

    Notice that there are limits on the number and types of instances that can run in a region. For example, there is a limit on the number of *Running On-Demand Standard...* instances that you can launch in this region. When launching instances,

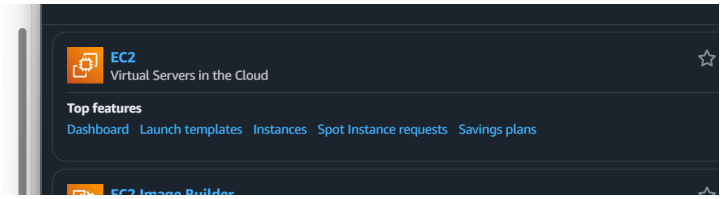


- Test Stop Protection

- choose instance

54. In the AWS Management Console, in the search box next to **Services**, search for and choose `EC2` to return to the EC2 console.

55. In left navigation pane, choose **Instances**.



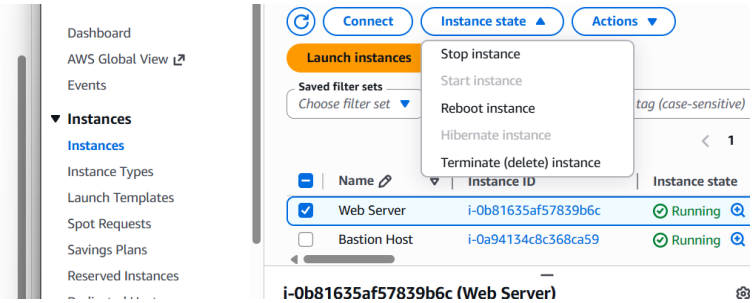
- stop instance

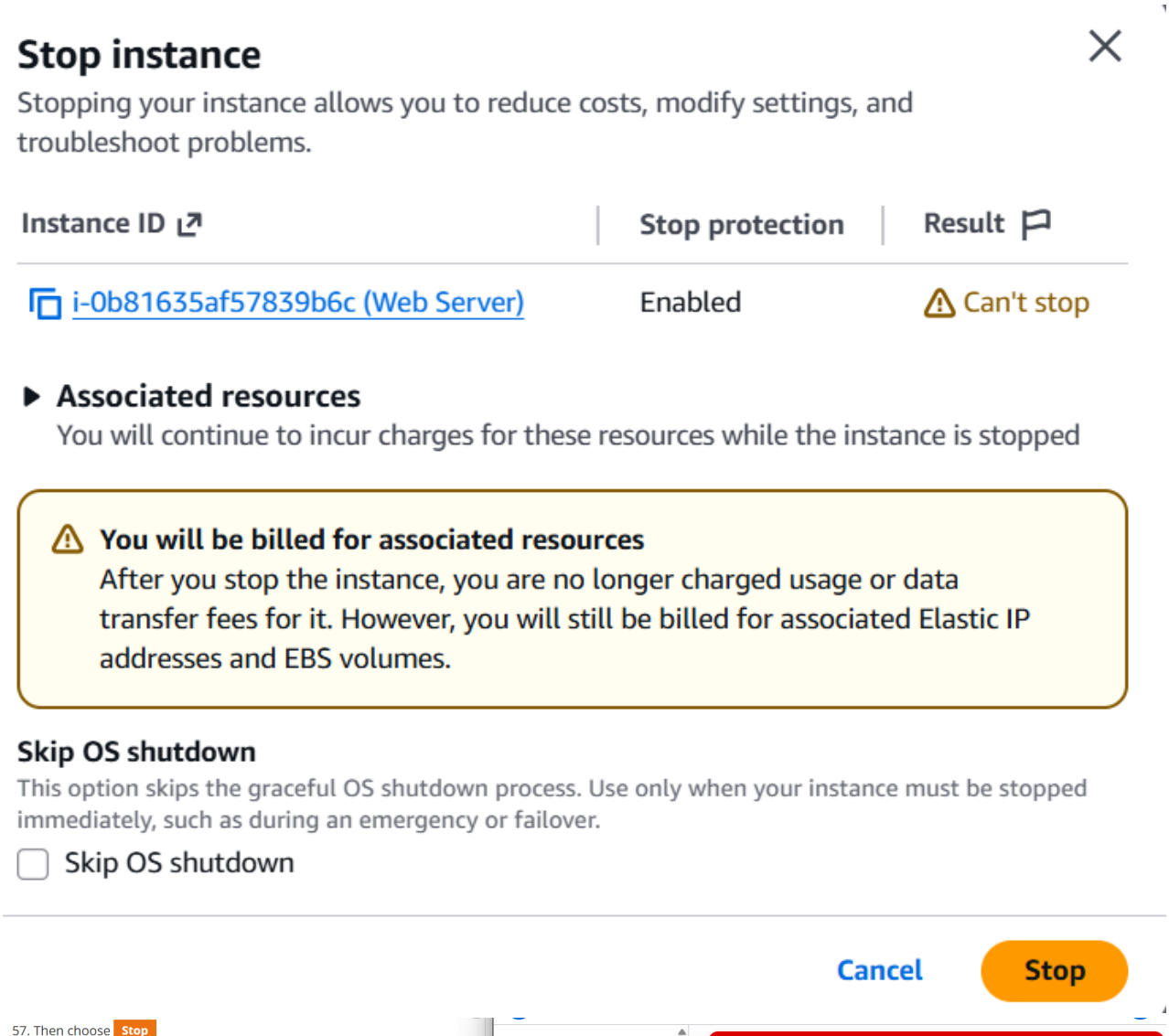
use *stop protection*.

54. In the AWS Management Console, in the search box next to **Services**, search for and choose `EC2` to return to the EC2 console.

55. In left navigation pane, choose **Instances**.

56. Select the **Web Server** instance and in the Instance state menu, select **Stop instance**.

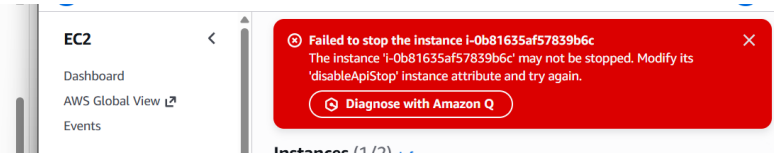




57. Then choose **Stop**

Note that there is a message that says: *Failed to stop the instance i-1234567xxx. The instance 'i-1234567xxx' may not be stopped. Modify its 'disableApiStop' instance attribute and try again.*

This shows that the stop protection that you enabled earlier in



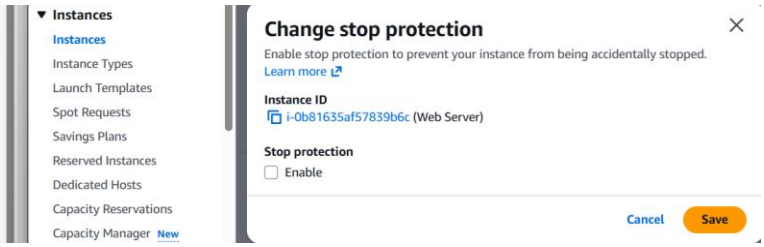
- Change stop protection

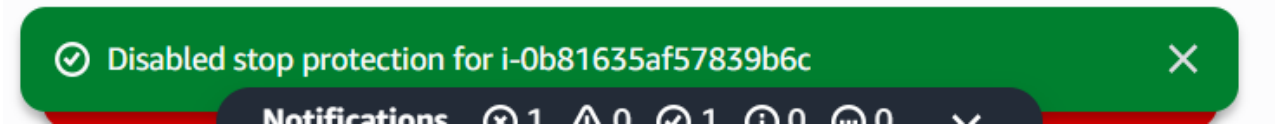
58. In the **Actions** menu, select **Instance settings** ▸ **Change stop protection**.

59. Remove the check next to ☐ **Enable**.

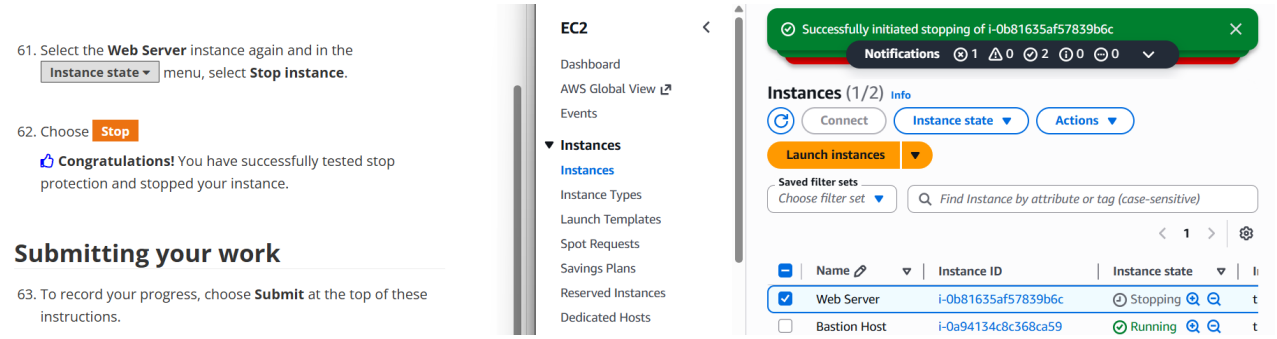
60. Choose **Save**

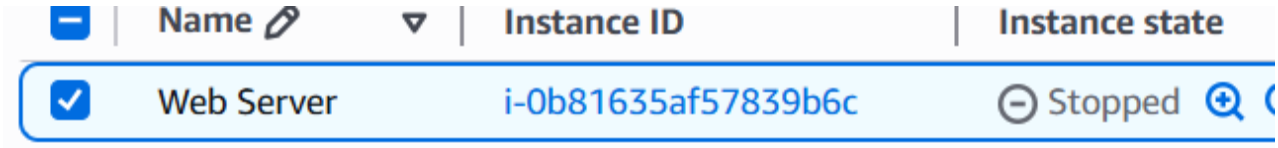
You can now stop the instance.





- Stop instance again

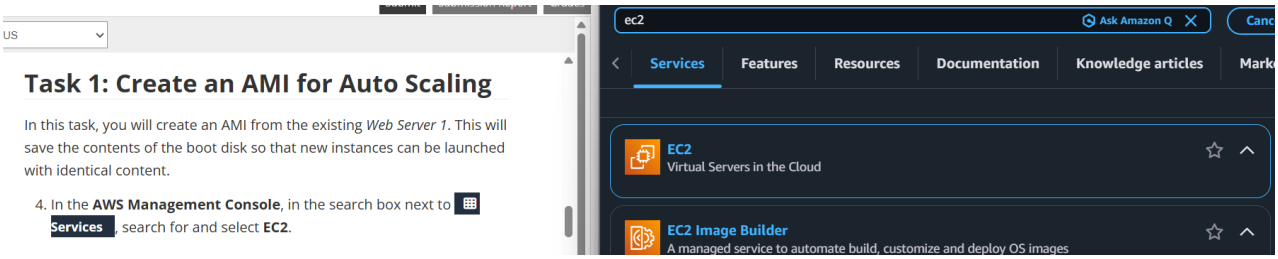




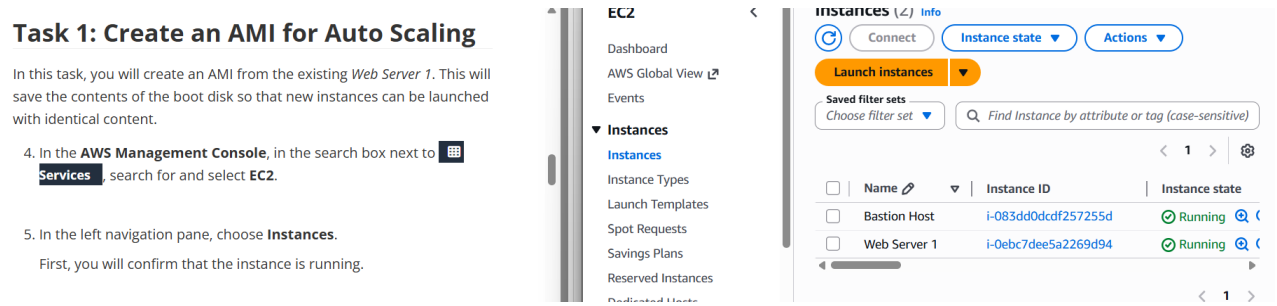
Task 3 (Access Module 10 Lab - 6 Scale & Load Balance your Architecture from AWS academy account) – Load Balancer & Auto Scaling Group setup and traffic network

- Create an AMI for Auto Scaling

- find EC2

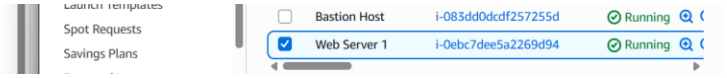


- Choose Instances



- Select web server 1

7. Select ☑ **Web Server 1**.



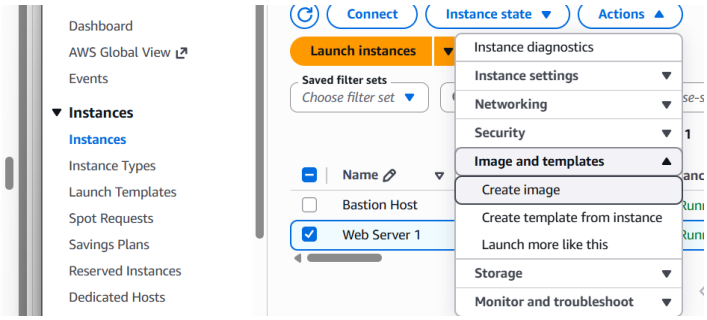
-Create image

7. Select ☑ **Web Server 1**.

8. In the Actions menu, choose **Image and templates** > **Create image**, then configure:
   - **Image name:** `WebServerAMI`
   - **Image description:** `Lab AMI for Web Server`

9. Choose **Create image**

   A confirmation banner displays the **AMI ID** for your new AMI.

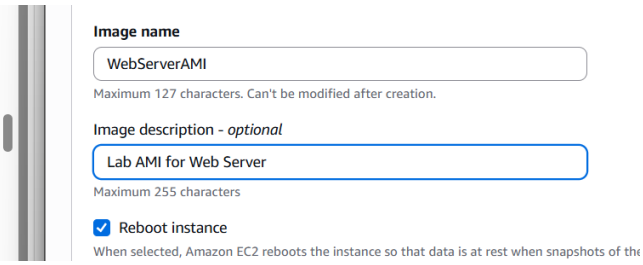


- Config

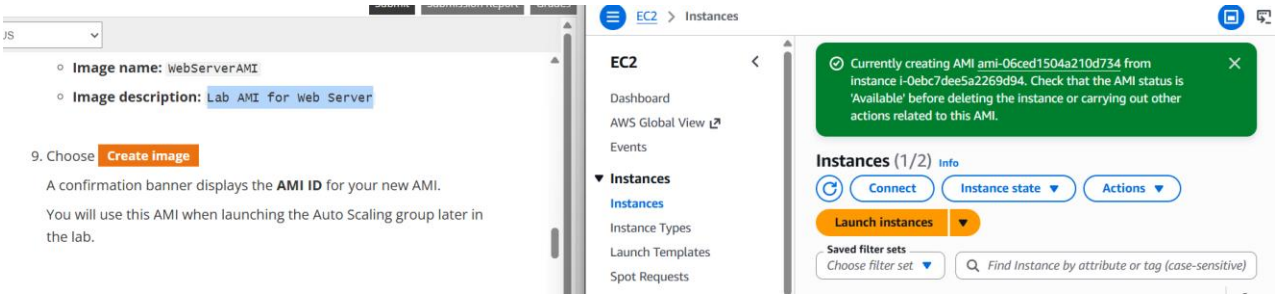
7. Select ☑ **Web Server 1**.

8. In the Actions menu, choose **Image and templates** > **Create image**, then configure:
   - **Image name:** `WebServerAMI`
   - **Image description:** `Lab AMI for Web Server`

9. Choose **Create image**



- Create



- Create a Load Balancer

- choose Target Groups

-US

the lab.

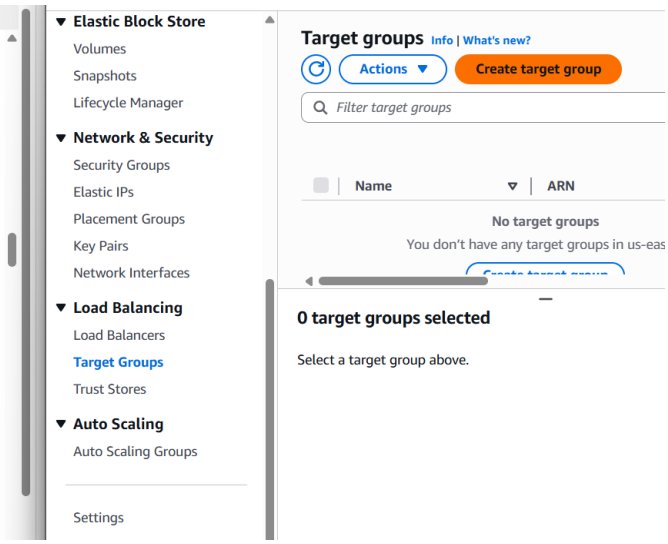
Task 2: Create a Load Balancer

In this task, you will first create a target group and then you will create a load balancer that can balance traffic across multiple EC2 instances and Availability Zones.

10. In the left navigation pane, choose **Target Groups**.

    Analysis: *Target Groups* define where to *send* traffic that comes into the Load Balancer. The Application Load Balancer can send traffic to multiple Target Groups based upon the URL of the incoming request, such as having requests from mobile apps going to a different set of servers. Your web application will use only one Target Group.

    - Choose **Create target group**
    - Choose a target type: **Instances**
    - **Target group name**, enter: `LabGroup`
    - Select **Lab VPC** from the **VPC** drop-down menu.



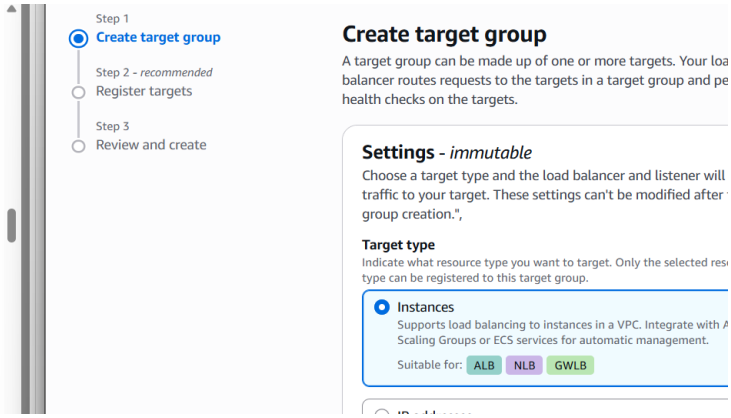
- create target group

s task, you will first create a target group and then you will create a balancer that can balance traffic across multiple EC2 instances and ability Zones.

n the left navigation pane, choose **Target Groups**.

nalysis: *Target Groups* define where to *send* traffic that comes into he Load Balancer. The Application Load Balancer can send traffic to nultiple Target Groups based upon the URL of the incoming request, uch as having requests from mobile apps going to a different set of ervers. Your web application will use only one Target Group.

- Choose **Create target group**
- Choose a target type: **Instances**
- **Target group name**, enter: `LabGroup`
- Select **Lab VPC** from the **VPC** drop-down menu.



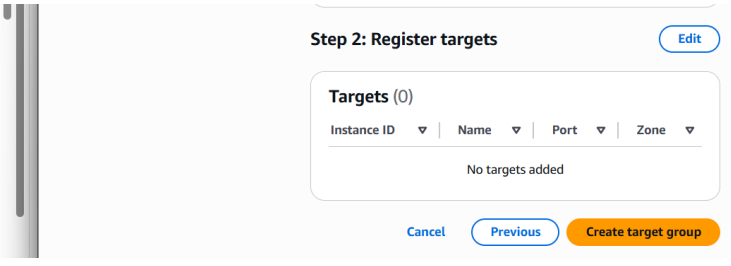
11. Choose **Next**. The **Register targets** screen appears.

    Note: *Targets* are the individual instances that will respond to requests from the Load Balancer.

    You do not have any web application instances yet, so you can skip this step.

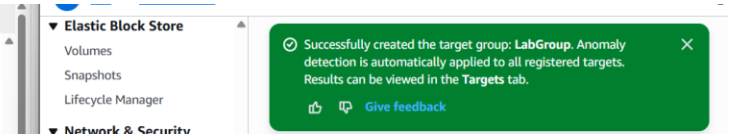
12. Review the settings and choose **Create target group**

13. In the left navigation pane, choose **Load Balancers**.

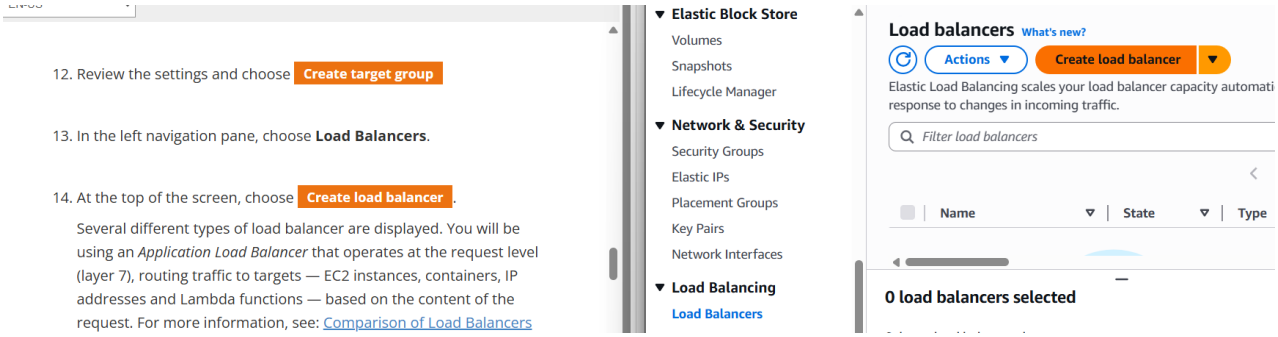


US

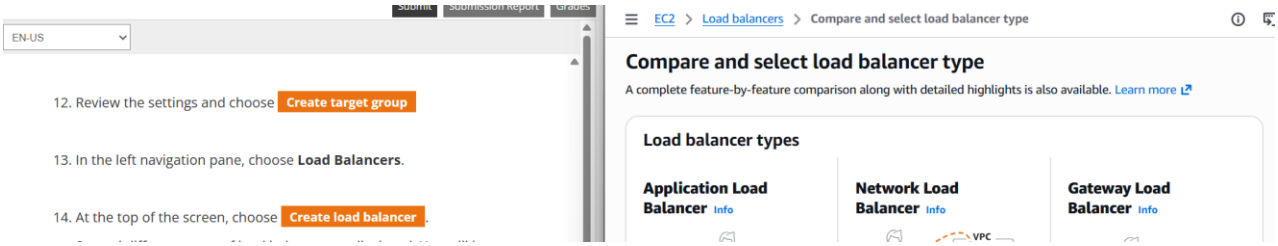
12. Review the settings and choose **Create target group**



- choose load balancer



- Create load balancer



- Choose application load balancer

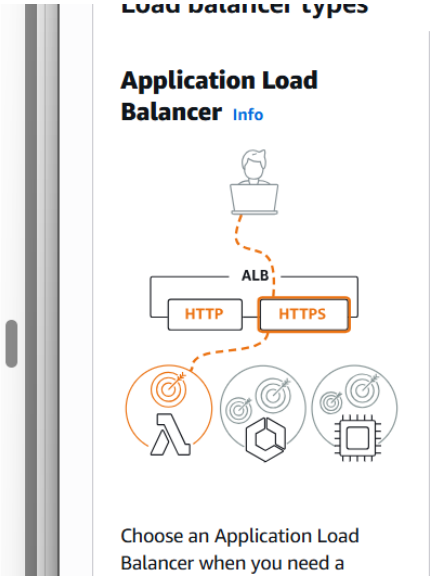
12. Review the settings and choose **Create target group**

13. In the left navigation pane, choose **Load Balancers**.

14. At the top of the screen, choose **Create load balancer**.

    Several different types of load balancer are displayed. You will be using an *Application Load Balancer* that operates at the request level (layer 7), routing traffic to targets — EC2 instances, containers, IP addresses and Lambda functions — based on the content of the request. For more information, see: Comparison of Load Balancers

15. Under **Application Load Balancer**, choose **Create**



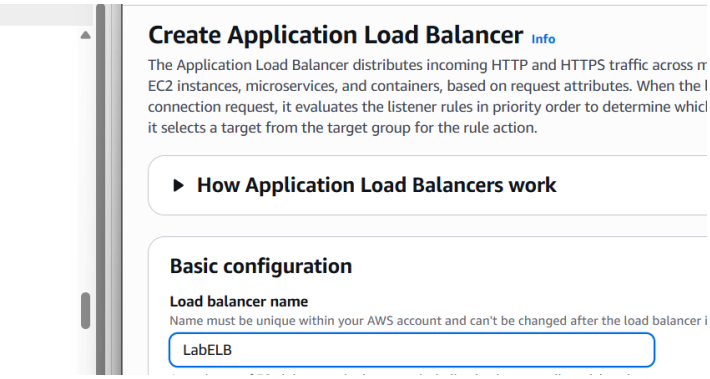
- Enter name

Several different types of load balancer are displayed. You will be using an *Application Load Balancer* that operates at the request level (layer 7), routing traffic to targets — EC2 instances, containers, IP addresses and Lambda functions — based on the content of the request. For more information, see: Comparison of Load Balancers

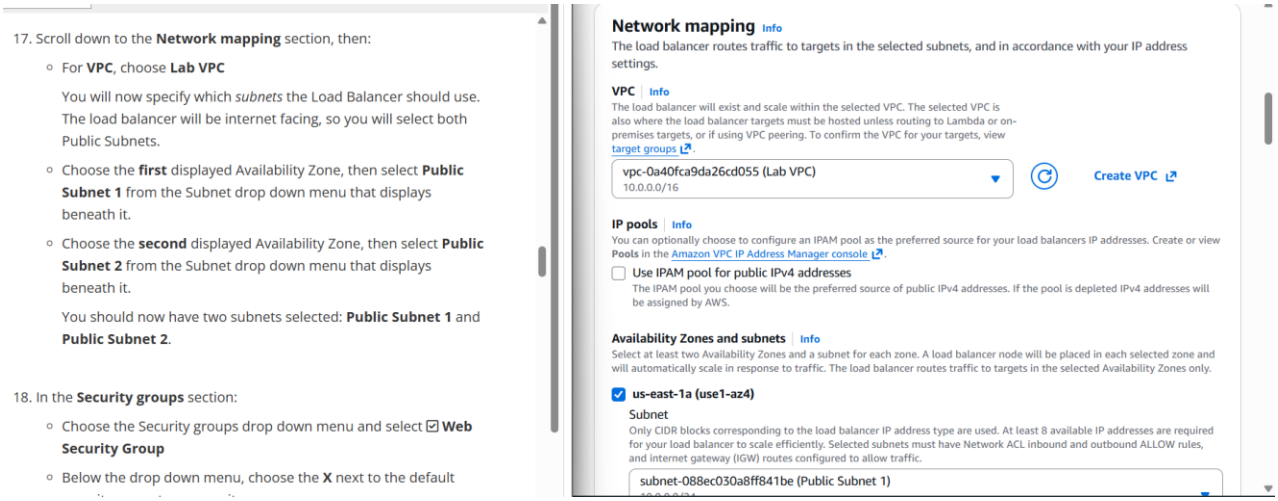
15. Under **Application Load Balancer**, choose Create

16. Under **Load balancer name**, enter: `LabELB`

17. Scroll down to the **Network mapping** section, then:



- network mapping

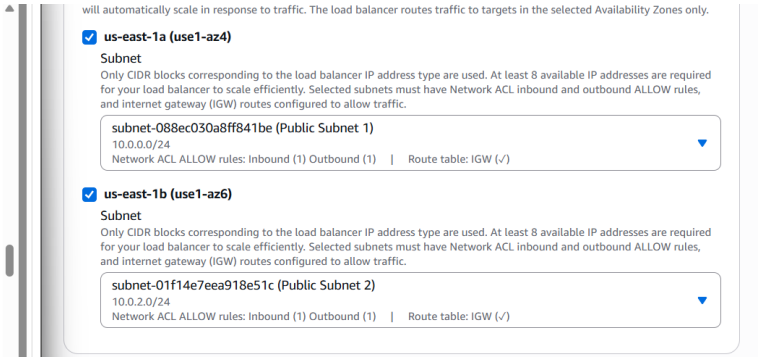


scroll down to the **Network mapping** section, then:

- For **VPC**, choose **Lab VPC**

  You will now specify which *subnets* the Load Balancer should use. The load balancer will be internet facing, so you will select both Public Subnets.
- Choose the **first** displayed Availability Zone, then select **Public Subnet 1** from the Subnet drop down menu that displays beneath it.
- Choose the **second** displayed Availability Zone, then select **Public Subnet 2** from the Subnet drop down menu that displays beneath it.

  You should now have two subnets selected: **Public Subnet 1** and **Public Subnet 2**.

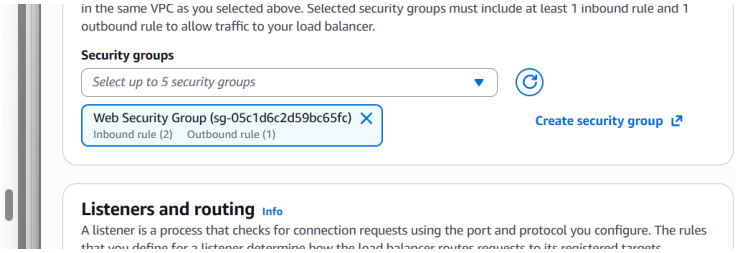


- Security group

18. In the **Security groups** section:

- Choose the Security groups drop down menu and select ☑ **Web Security Group**
- Below the drop down menu, choose the **X** next to the default security group to remove it.

  The **Web Security Group** security group should now be the only one that appears.



- For the Listener HTTP:80 row, set the Default action to forward to LabGroup

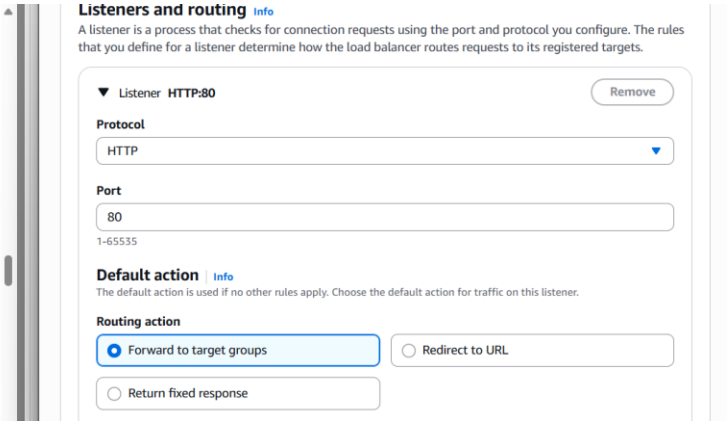
18. In the **Security groups** section:
    - Choose the Security groups drop down menu and select ☑ **Web Security Group**
    - Below the drop down menu, choose the **X** next to the default security group to remove it.

      The **Web Security Group** security group should now be the only one that appears.

19. For the Listener HTTP:80 row, set the Default action to forward to **LabGroup**.

20. Scroll to the bottom and choose **Create load balancer**

    The load balancer is successfully created.
    - Choose **View load balancer**



- Create load balancer

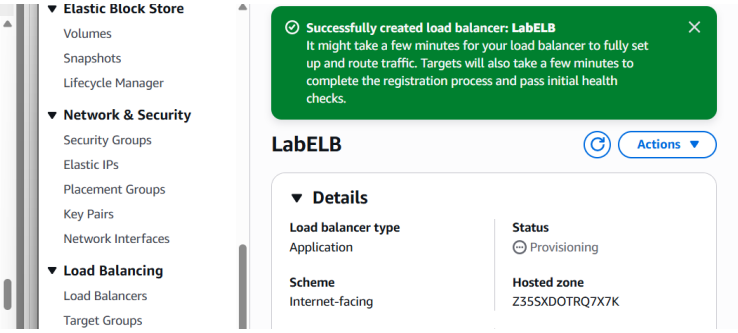
one that appears.

19. For the Listener HTTP:80 row, set the Default action to forward to **LabGroup**.

20. Scroll to the bottom and choose **Create load balancer**

    The load balancer is successfully created.
    - Choose **View load balancer**

      The load balancer will show a state of *provisioning*. There is no need to wait until it is ready. Please continue with the next task.



- Create a Launch Template and an Auto Scaling Group

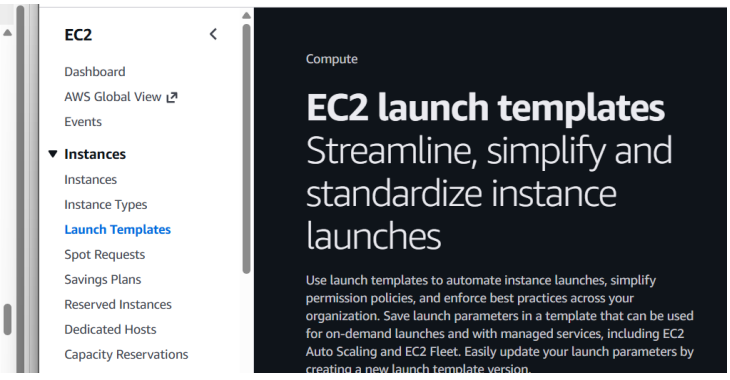
- choose Launch Templates

need to wait until it is ready. Please continue with the next task.

Task 3: Create a Launch Template and an Auto Scaling Group

In this task, you will create a *launch template* for your Auto Scaling group. A launch template is a template that an Auto Scaling group uses to launch EC2 instances. When you create a launch template, you specify information for the instances such as the AMI, the instance type, a key pair, and security group.

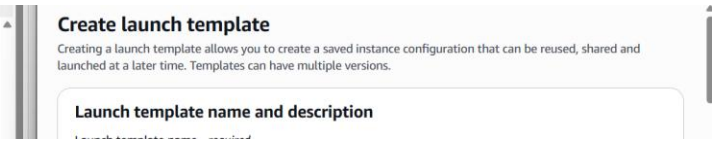
21. In the left navigation pane, choose **Launch Templates**.



- create launch template

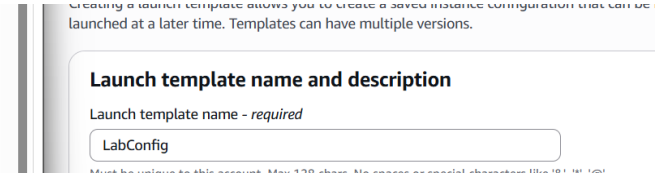
21. In the left navigation pane, choose **Launch Templates**.

22. Choose **Create launch template**



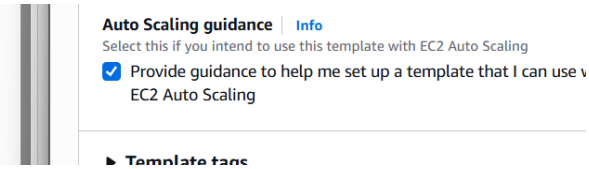
- template name

23. Configure the launch template settings and create it:
    - **Launch template name:** `LabConfig`
    - Under **Auto Scaling guidance**, select ☑ *Provide guidance to help me set up a template that I can use with EC2 Auto Scaling*



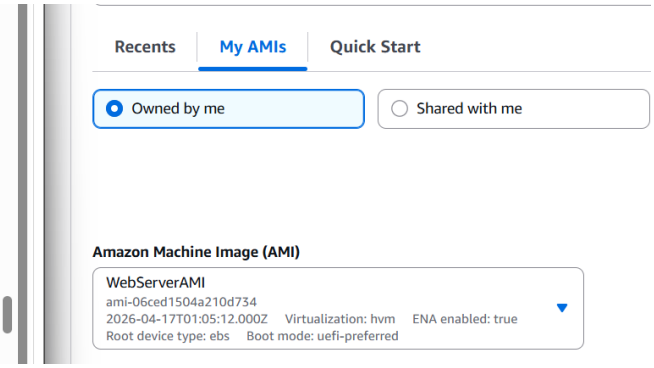
- select Provide guidance

- ~~**Launch template name:** `LabConfig`~~
- Under **Auto Scaling guidance**, select ☑ *Provide guidance to help me set up a template that I can use with EC2 Auto Scaling*
- In the Application and OS Images (Amazon Machine Image) area, choose *My AMIs*.



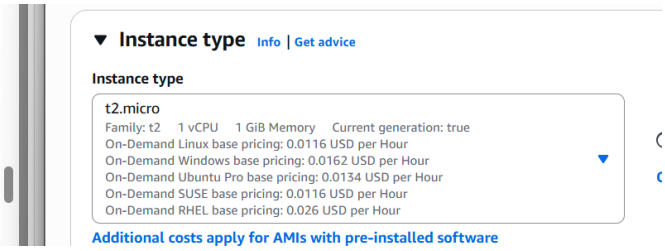
- choose Web Server AMI

. Configure the launch template settings and create it:
- **Launch template name:** `LabConfig`
- Under **Auto Scaling guidance**, select ☑ *Provide guidance to help me set up a template that I can use with EC2 Auto Scaling*
- In the Application and OS Images (Amazon Machine Image) area, choose *My AMIs*.
- **Amazon Machine Image (AMI)**: choose *Web Server AMI*
- **Instance type:** choose *t2.micro*
- **Key pair name**: choose *vockey*
- **Firewall (security groups)**: choose *Select existing security group*



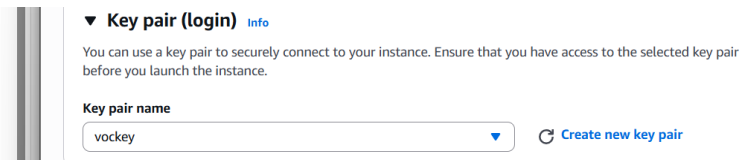
- instance type

me set up a template that I can use with EC2 Auto Scaling
- In the Application and OS Images (Amazon Machine Image) area, choose *My AMIs*.
- **Amazon Machine Image (AMI)**: choose *Web Server AMI*
- **Instance type:** choose *t2.micro*
- **Key pair name**: choose *vockey*
- **Firewall (security groups)**: choose *Select existing security group*
- **Security groups**: choose ☑ *Web Security Group*



- key pair

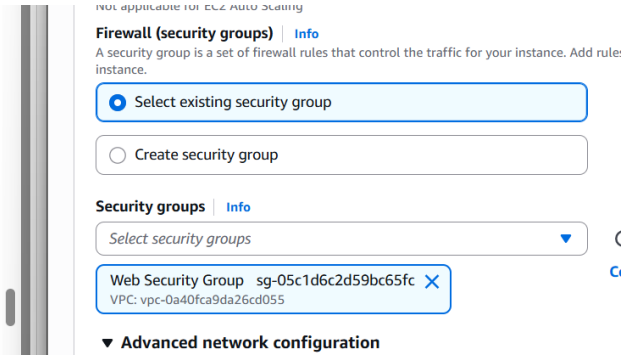
- **Amazon Machine Image (AMI)**: choose *Web Server AMI*
- **Instance type:** choose *t2.micro*
- **Key pair name**: choose *vockey*
- **Firewall (security groups)**: choose *Select existing security group*
- **Security groups**: choose ☑ *Web Security Group*



- Firewall (security groups)

me set up a template that I can use with EC2 Auto Scaling

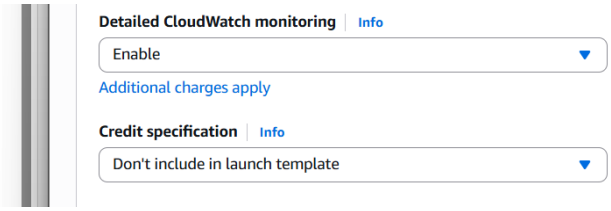
- In the Application and OS Images (Amazon Machine Image) area, choose *My AMIs*.
- **Amazon Machine Image (AMI)**: choose *Web Server AMI*
- **Instance type:** choose *t2.micro*
- **Key pair name**: choose *vockey*
- **Firewall (security groups)**: choose *Select existing security group*
- **Security groups**: choose ☑ *Web Security Group*
- Scroll down to the **Advanced details** area and expand it.
- Scroll down to the **Detailed CloudWatch monitoring** setting. Select ☑ *Enable*



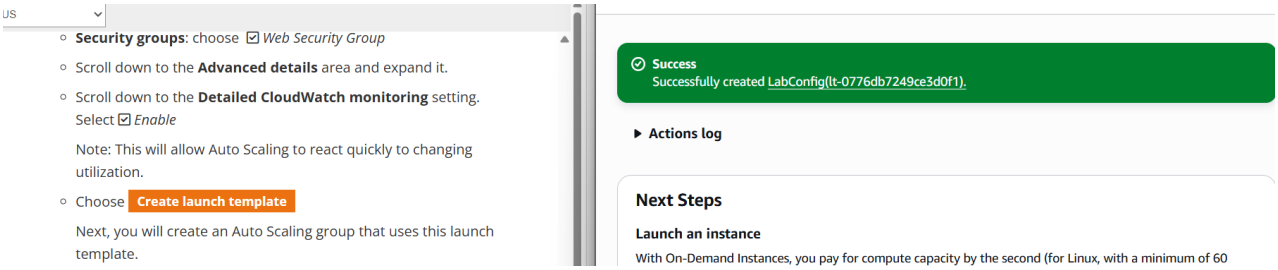
- Advanced details

- **Security groups**: choose ☑ *Web Security Group*
- Scroll down to the **Advanced details** area and expand it.
- Scroll down to the **Detailed CloudWatch monitoring** setting. Select ☑ *Enable*

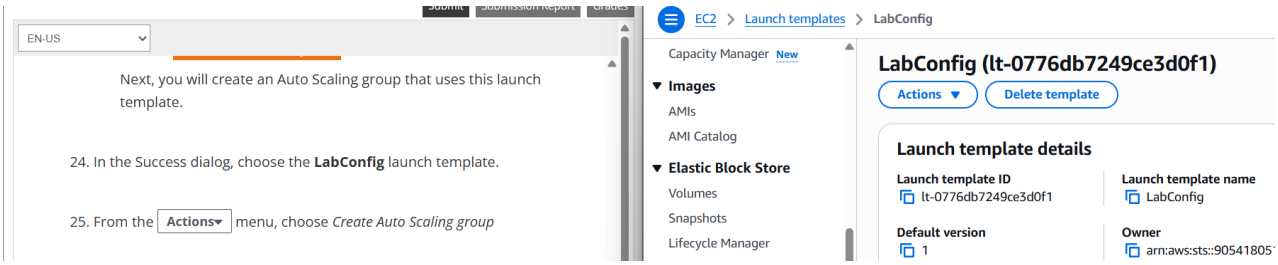
  Note: This will allow Auto Scaling to react quickly to changing utilization.



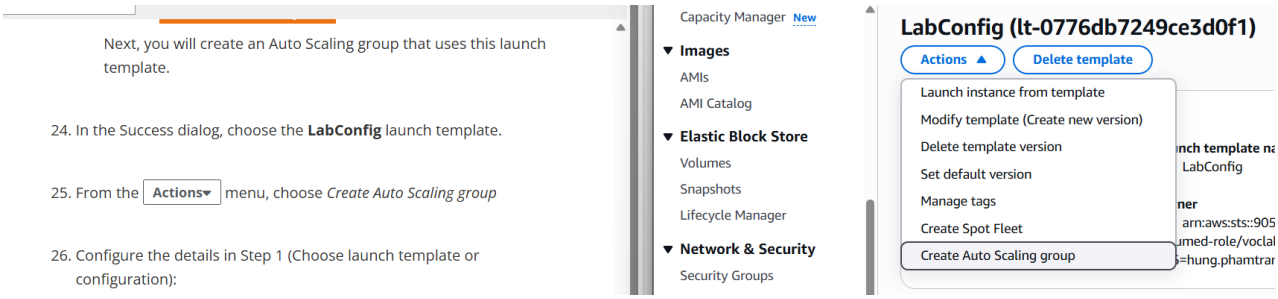
- Create launch template



- choose lab config

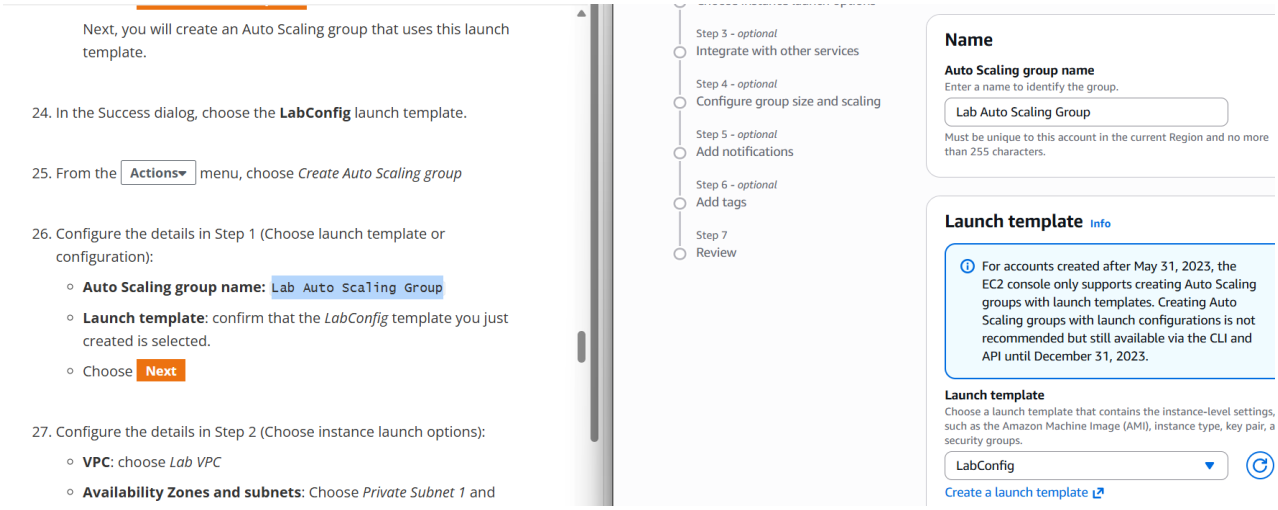


- Choose Create Auto Scaling group



-Step 1 (Choose launch template or configuration

- **Auto Scaling group name:** `Lab Auto Scaling Group`
- **Launch template**: confirm that the *LabConfig* template you just created is selected.
- Choose **Next**

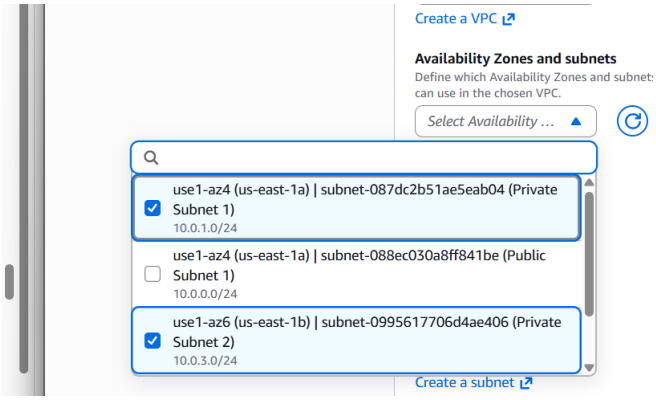


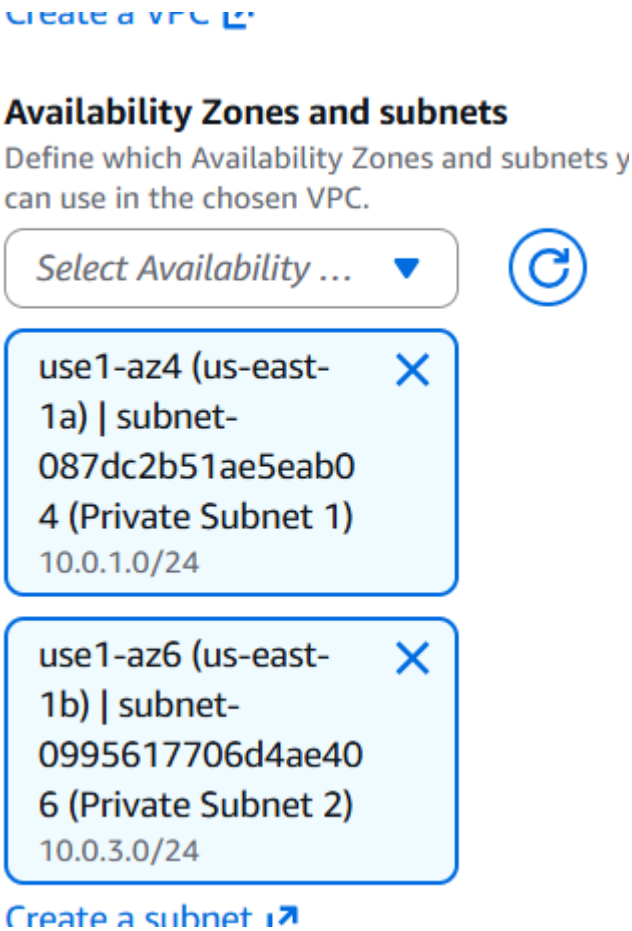
- Step 2 (Choose instance launch options)

- **VPC**: choose *Lab VPC*
- **Availability Zones and subnets**: Choose *Private Subnet 1* and then choose *Private Subnet 2*.
- Choose **Next**

27. Configure the details in Step 2 (Choose instance launch options):
    - **VPC**: choose *Lab VPC*
    - **Availability Zones and subnets**: Choose *Private Subnet 1* and then choose *Private Subnet 2*.
    - Choose **Next**

28. Configure the details in Step 3 (Configure advanced options):
    - Choose **Attach to an existing load balancer**
        - **Existing load balancer target groups**: select *LabGroup*.
    - In the **Additional settings** pane:
        - Select ☑ **Enable group metrics collection within**





- Step 3 (Configure advanced options)

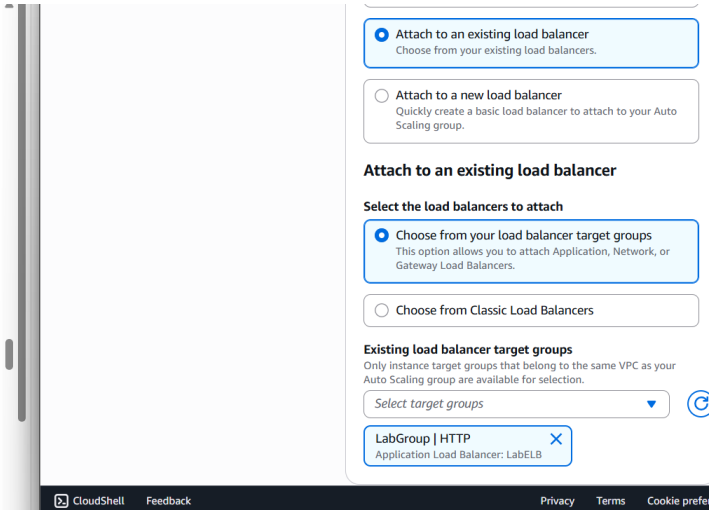
- Choose **Attach to an existing load balancer**
    - **Existing load balancer target groups**: select *LabGroup*.
- In the **Additional settings** pane:
    - Select **Enable group metrics collection within CloudWatch**

28. Configure the details in Step 3 (Configure advanced options):
    - Choose **Attach to an existing load balancer**
        - **Existing load balancer target groups**: select *LabGroup*.
    - In the **Additional settings** pane:
        - Select ☑ **Enable group metrics collection within CloudWatch**

      This will capture metrics at 1-minute intervals, which allows Auto Scaling to react quickly to changing usage patterns.
    - Choose **Next**

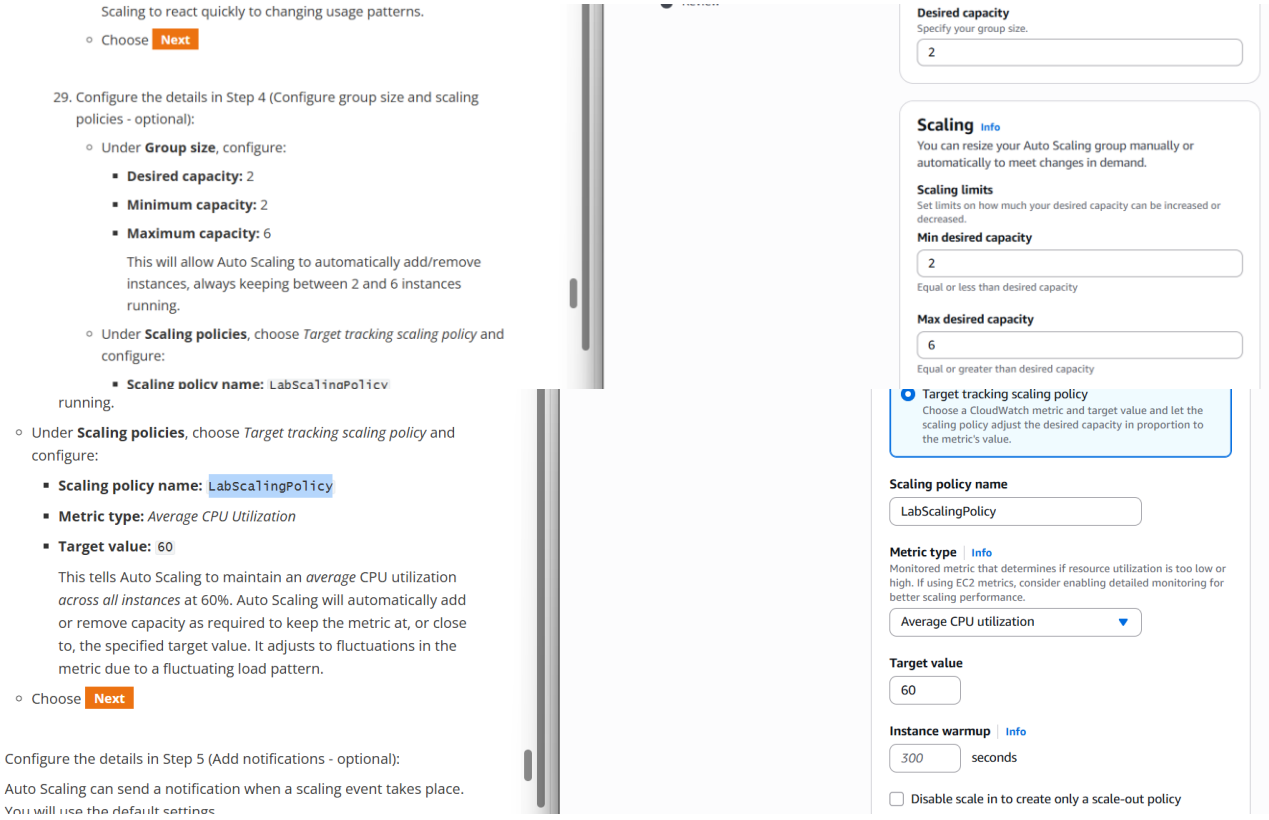
29. Configure the details in Step 4 (Configure group size and scaling policies - optional):
    - Under **Group size**, configure:
        - **Desired capacity:** 2
        - **Minimum capacity:** 2
        - **Maximum capacity:** 6

      This will allow Auto Scaling to automatically add/remove



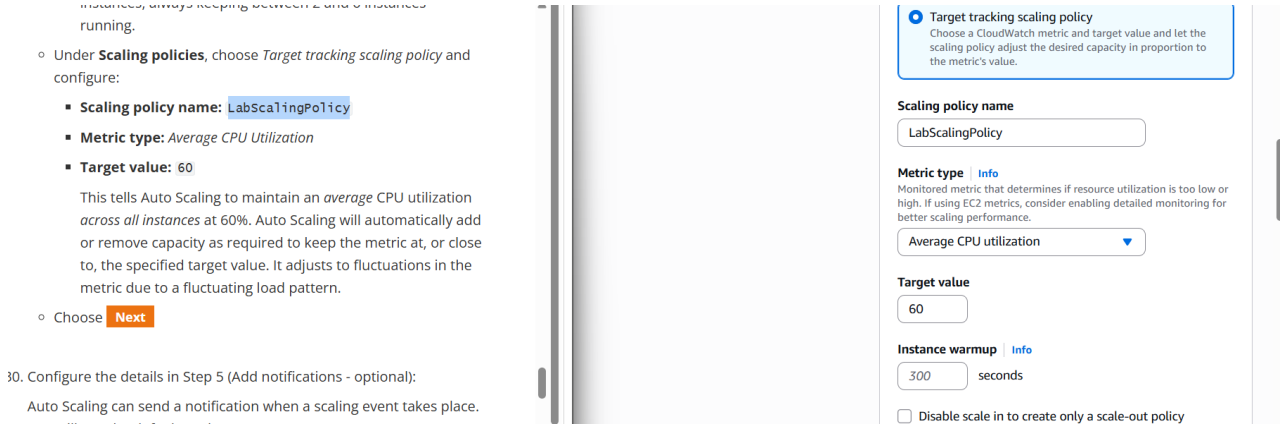
Configure the details in Step 4 (Configure group size and scaling policies - optional):

- Under **Group size**, configure:
- **Desired capacity:** 2
- **Minimum capacity:** 2
- **Maximum capacity:** 6

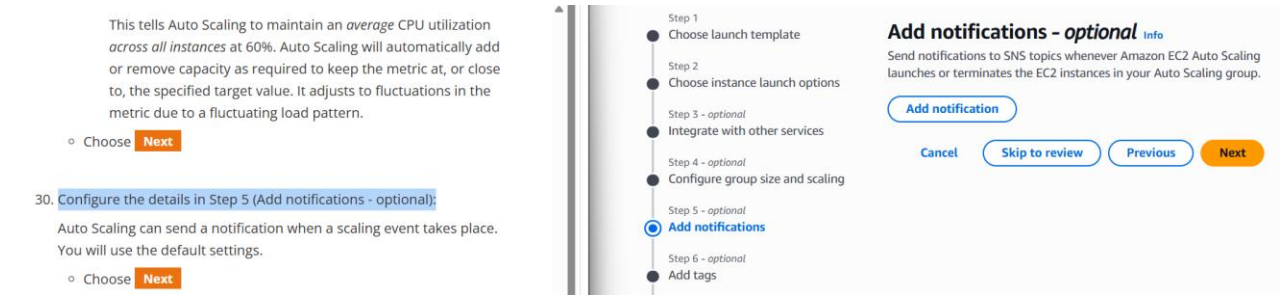


- Under **Scaling policies**, choose *Target tracking scaling policy* and configure:

- **Scaling policy name:** `LabScalingPolicy`
- **Metric type:** *Average CPU Utilization*
- **Target value:** `60`



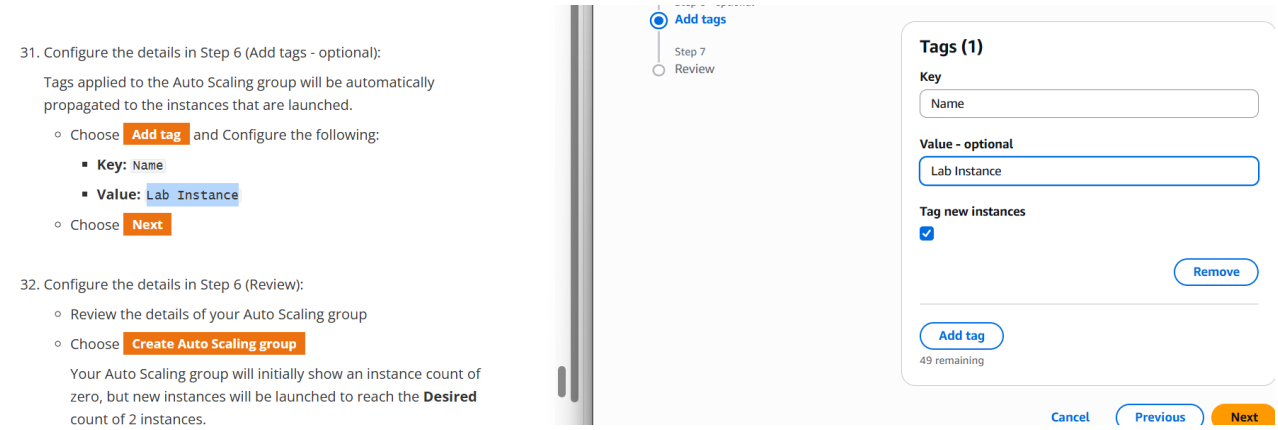
- Configure the details in Step 5 (Add notifications - optional):



- Configure the details in Step 6 (Add tags - optional):

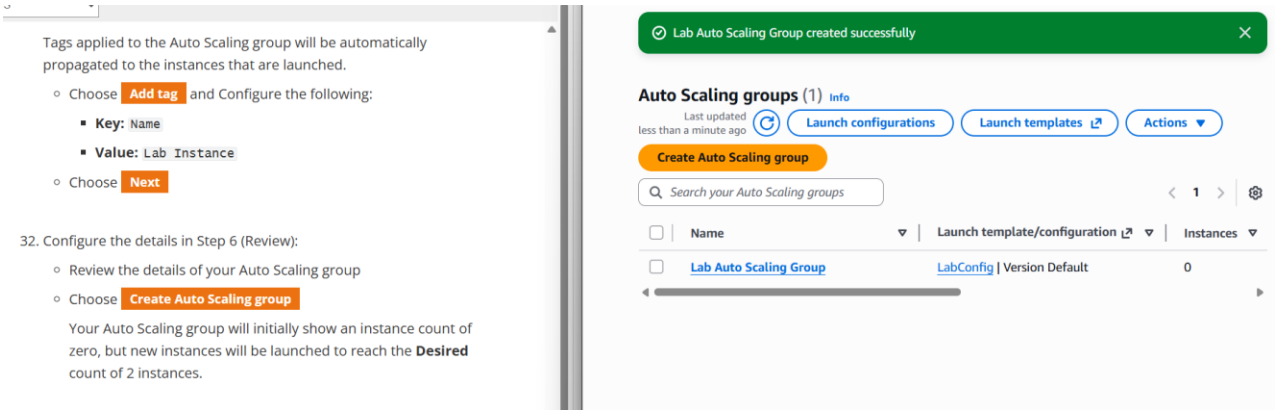
Tags applied to the Auto Scaling group will be automatically propagated to the instances that are launched.

- Choose **Add tag** and Configure the following:
- **Key:** `Name`
- **Value:** `Lab Instance`



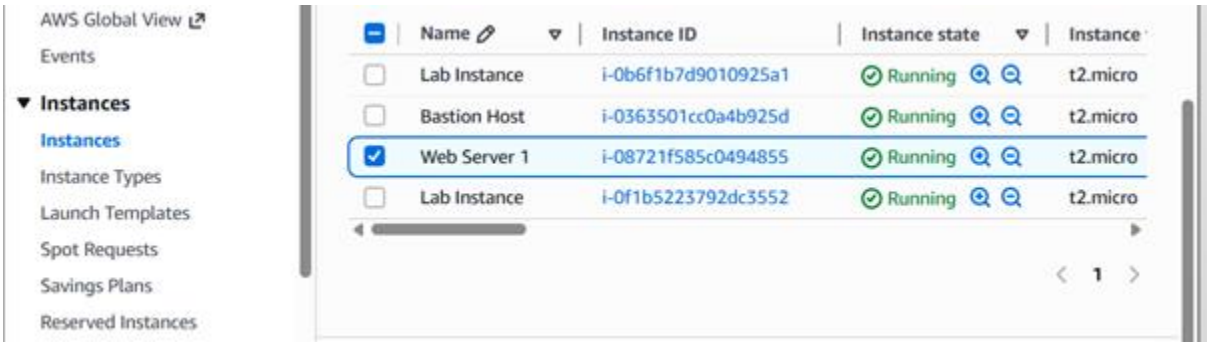
21. **Configure the details in Step 6 (Review):**
   - Review the details of your Auto Scaling group
   - Choose **Create Auto Scaling group**

   Your Auto Scaling group will initially show an instance count of zero, but new instances will be launched to reach the **Desired** count of 2 instances.

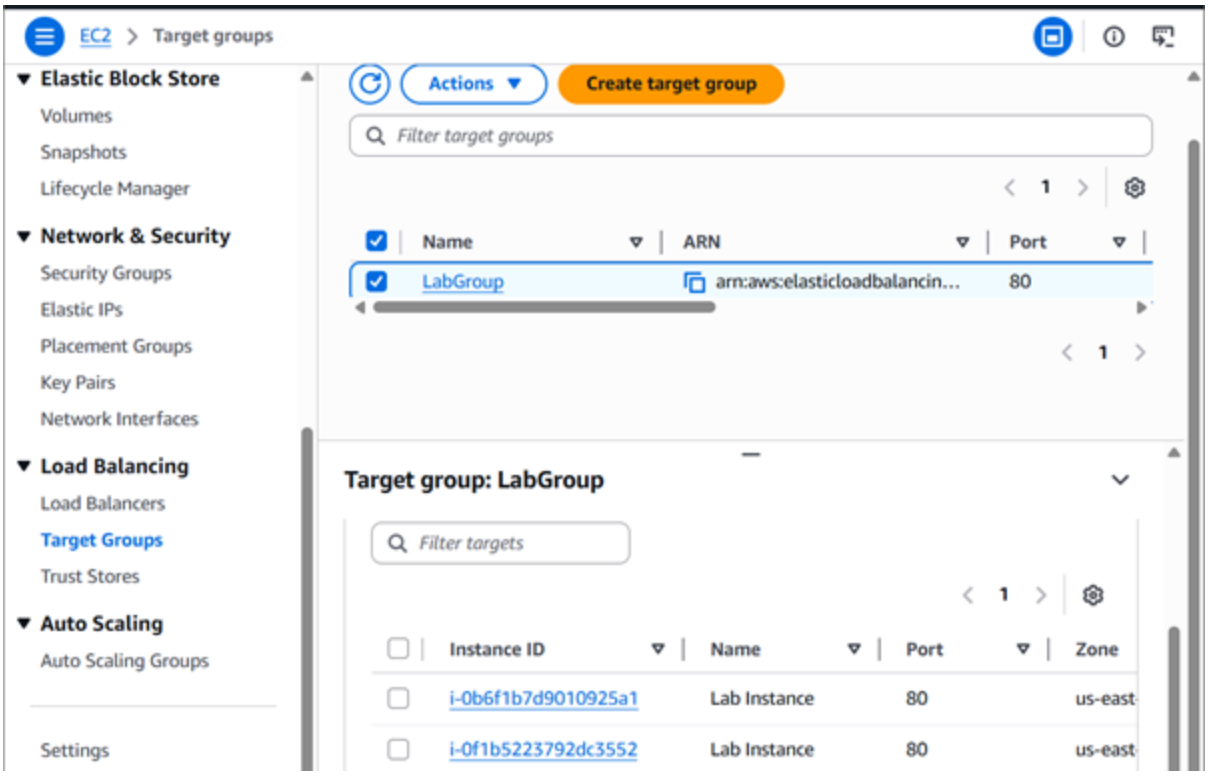


-Verify that Load Balancing is Working

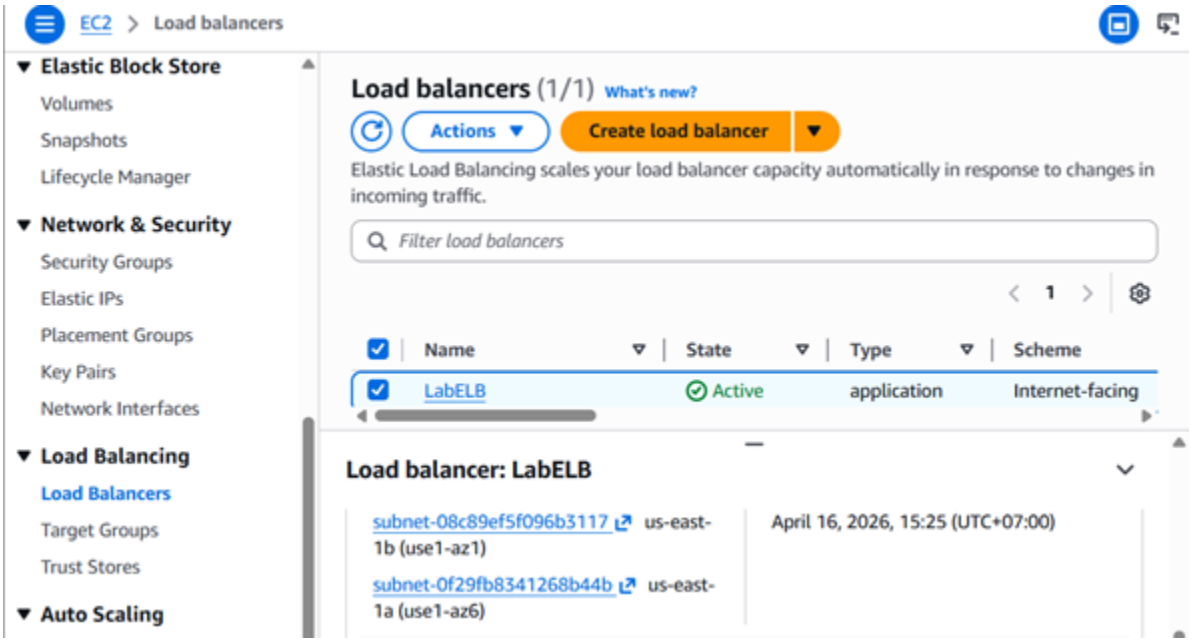
choose Instances



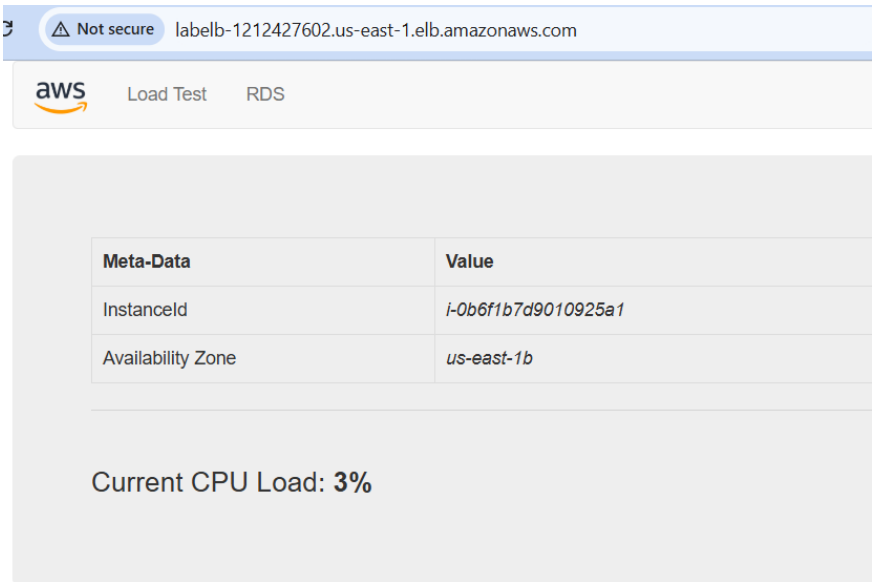
choose Target Groups



In the left navigation pane, choose Load Balancers



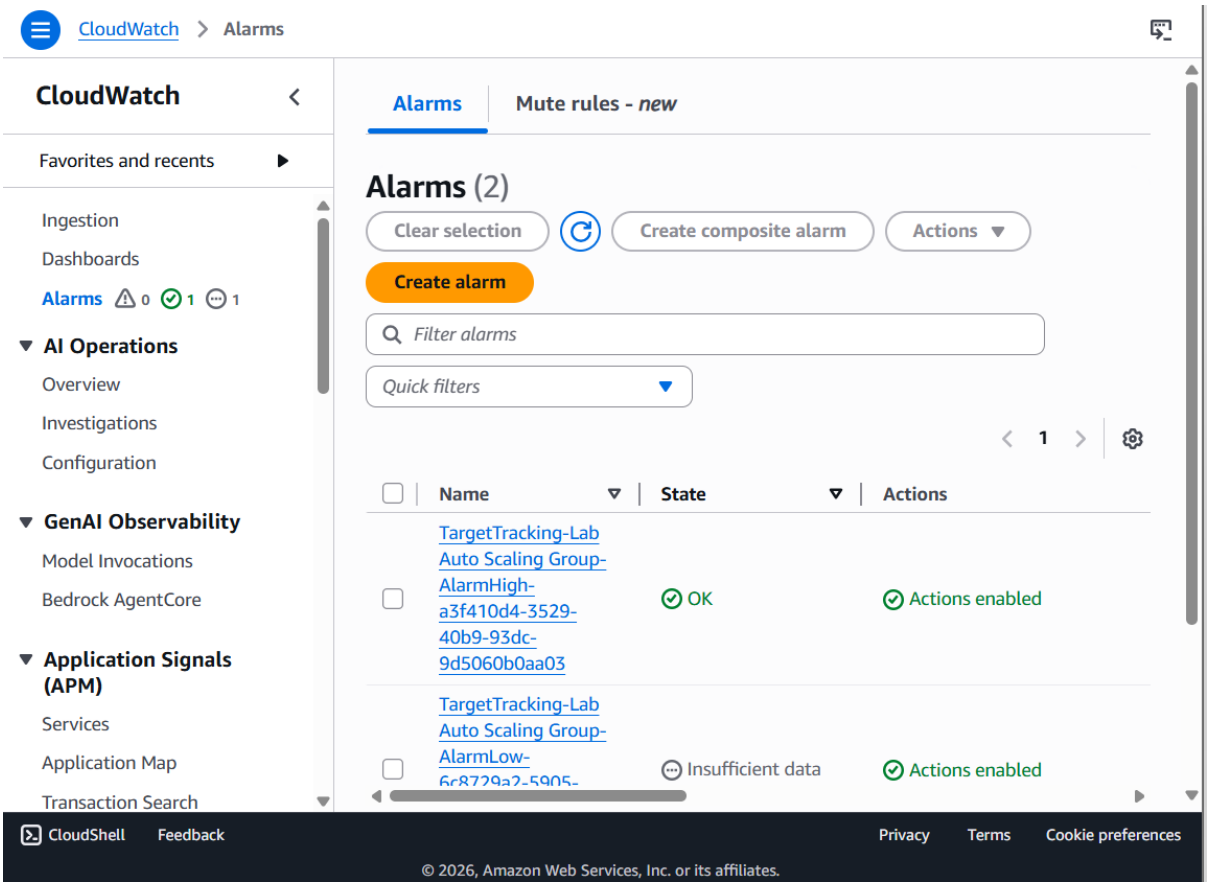
Open a new web browser tab, paste the DNS Name you just copied, and press Enter

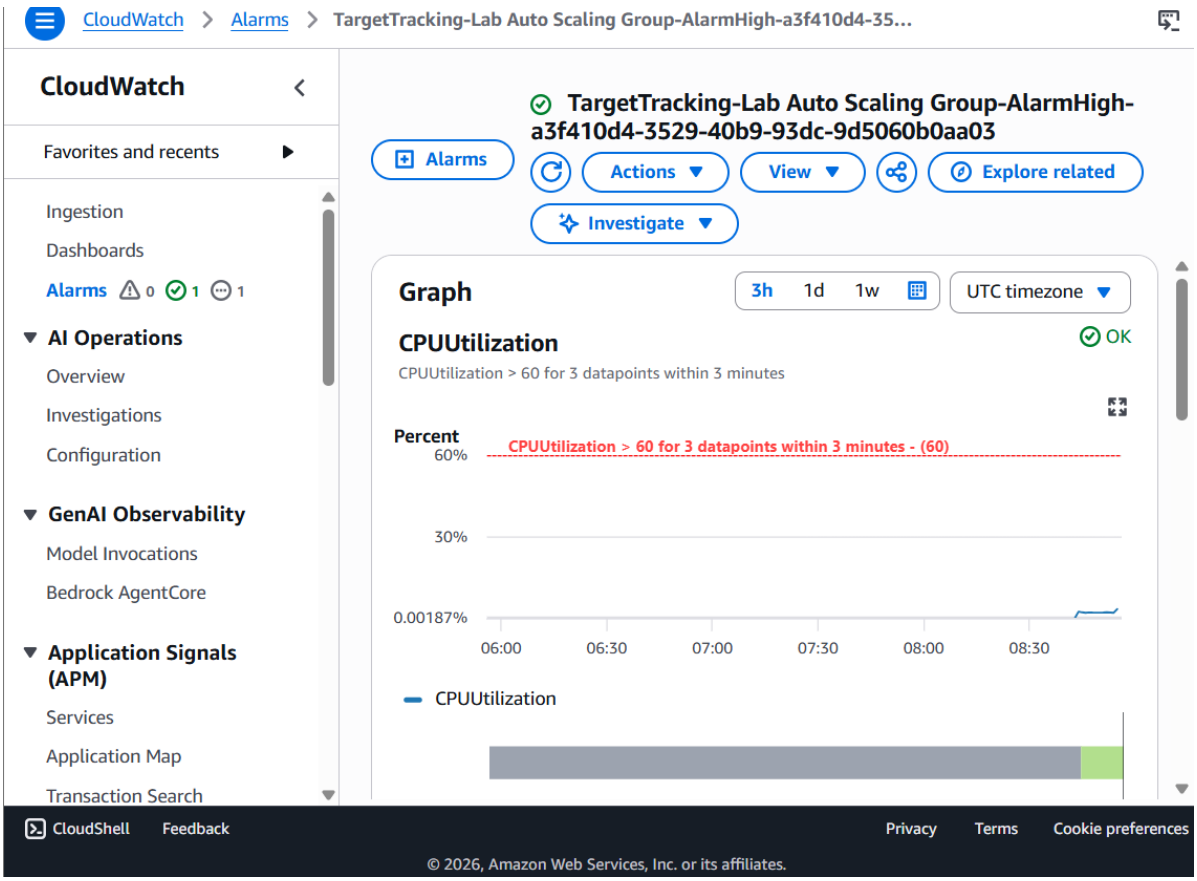


Test Auto Scaling

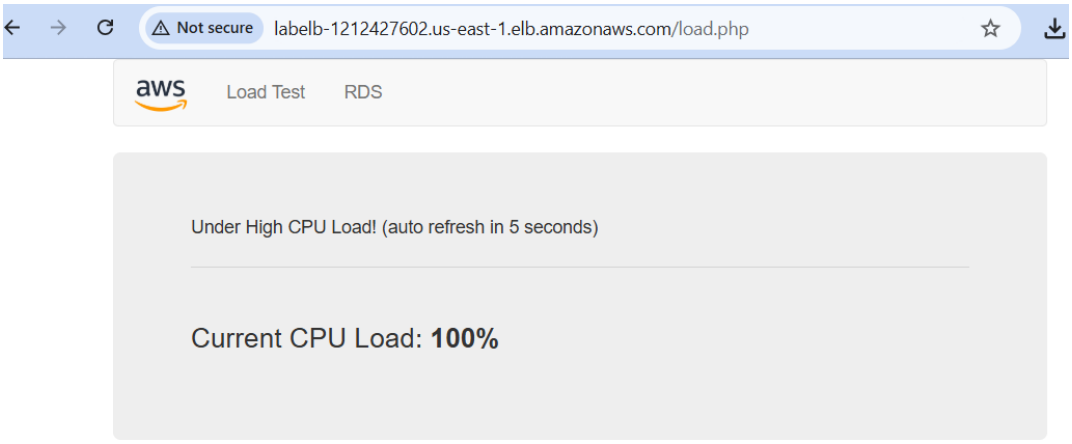
created an Auto Scaling group with a minimum of two instances and a maximum of six instances. Currently two instances are running because the minimum size is two and the group is currently not under any load. You will now increase the load to cause Auto Scaling to add additional instances

In the left navigation pane, choose All alarms

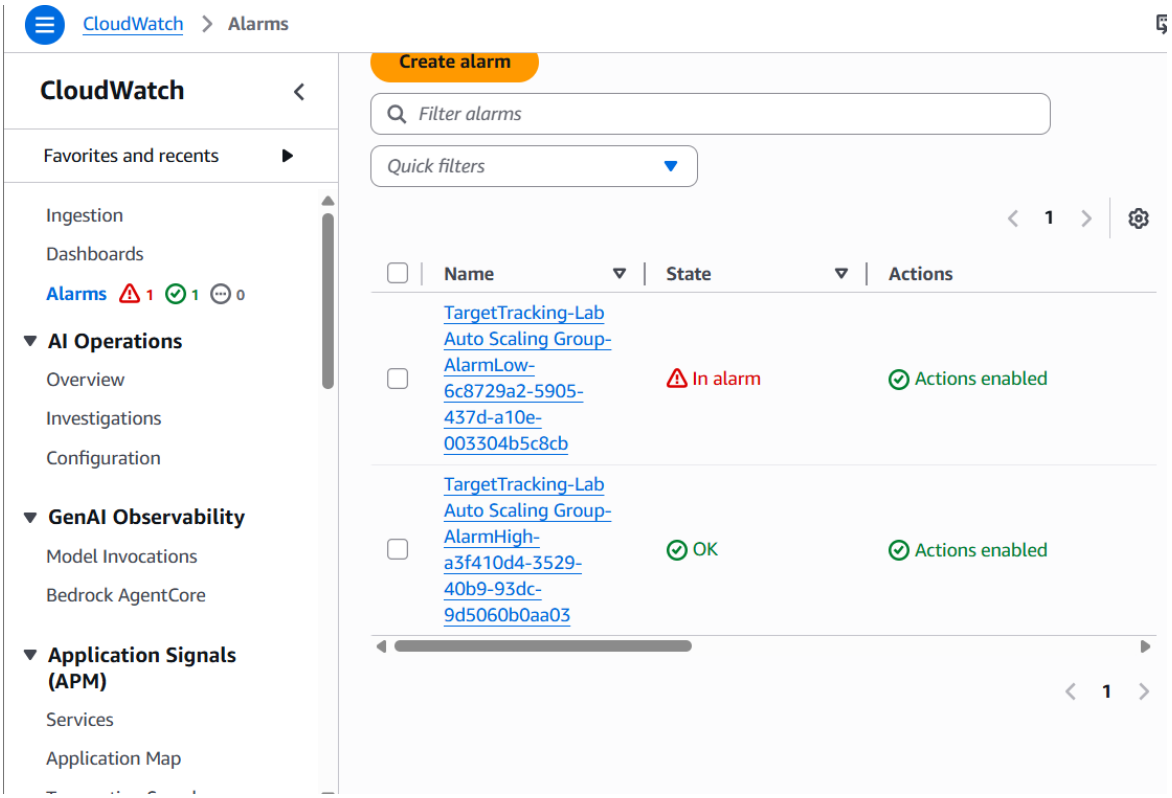

CloudWatch > Alarms
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms 0 1 1
AI Operations
Overview
Investigations
Configuration
GenAI Observability
Model Invocations
Bedrock AgentCore
Application Signals (APM)
Services
Application Map
Transaction Search
Alarms
Mute rules - new
Alarms (2)
Clear selection
Create composite alarm
Actions
Create alarm
Filter alarms
Quick filters
1
Name
State
Actions
TargetTracking-Lab Auto Scaling Group-AlarmHigh-a3f410d4-3529-40b9-93dc-9d5060b0aa03
OK
Actions enabled
TargetTracking-Lab Auto Scaling Group-AlarmLow-6c8729a2-5905-
Insufficient data
Actions enabled
CloudShell
Feedback
Privacy
Terms
Cookie preferences
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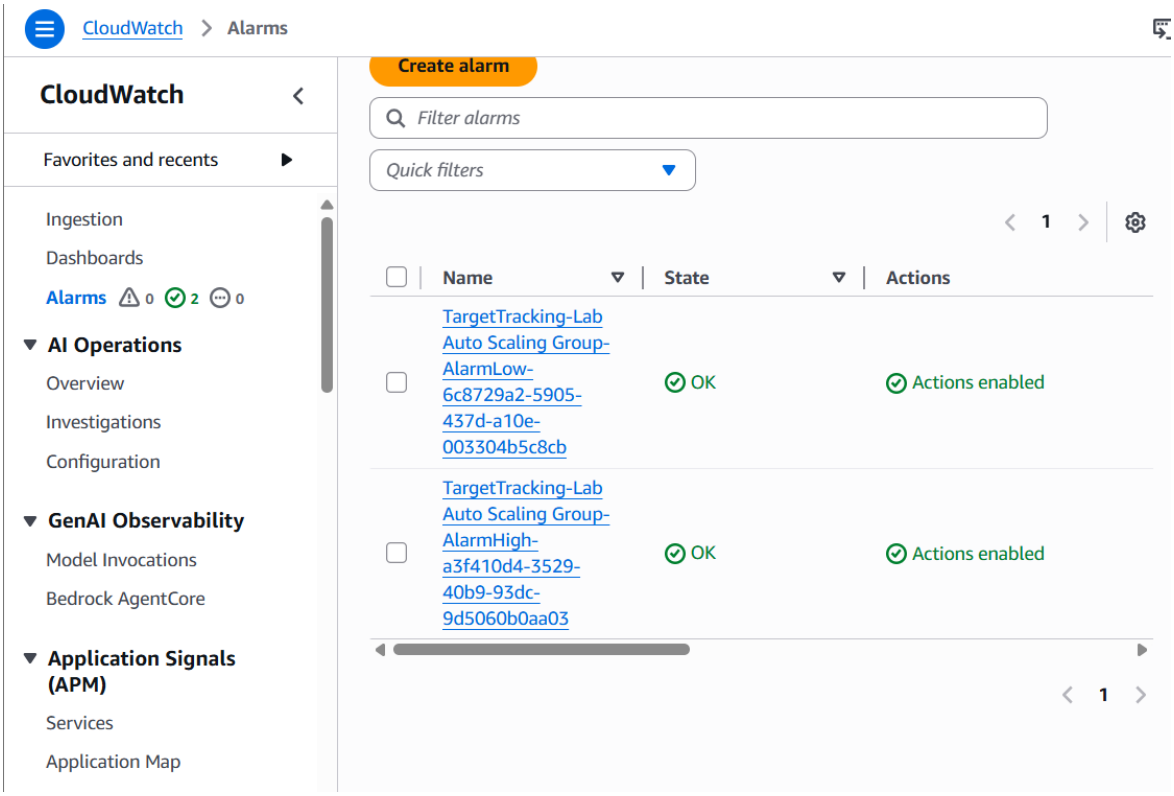

CloudWatch > Alarms > TargetTracking-Lab Auto Scaling Group-AlarmHigh-a3f410d4-35...
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms 0 1 1
AI Operations
Overview
Investigations
Configuration
GenAI Observability
Model Invocations
Bedrock AgentCore
Application Signals (APM)
Services
Application Map
Transaction Search
TargetTracking-Lab Auto Scaling Group-AlarmHigh-a3f410d4-3529-40b9-93dc-9d5060b0aa03
Alarms
Actions
View
Explore related
Investigate
Graph
3h
1d
1w
UTC timezone
CPUUtilization
OK
CPUUtilization > 60 for 3 datapoints within 3 minutes
Percent
60%
CPUUtilization > 60 for 3 datapoints within 3 minutes - (60)
30%
0.00187%
06:00
06:30
07:00
07:30
08:00
08:30
CPUUtilization
CloudShell
Feedback
Privacy
Terms
Cookie preferences
© 2026, Amazon Web Services, Inc. or its affiliates.

Return to browser tab with the CloudWatch console



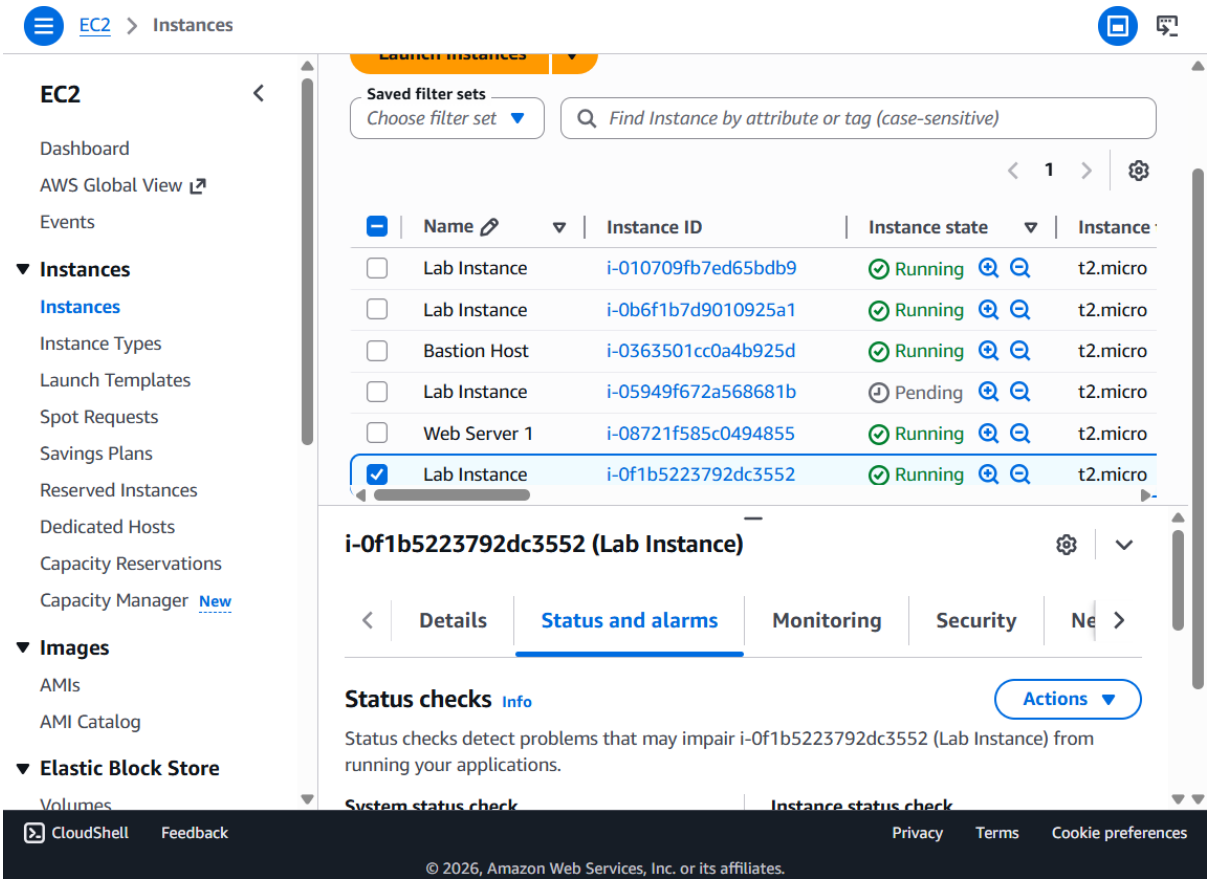
Wait until the AlarmHigh alarm enters the In alarm state

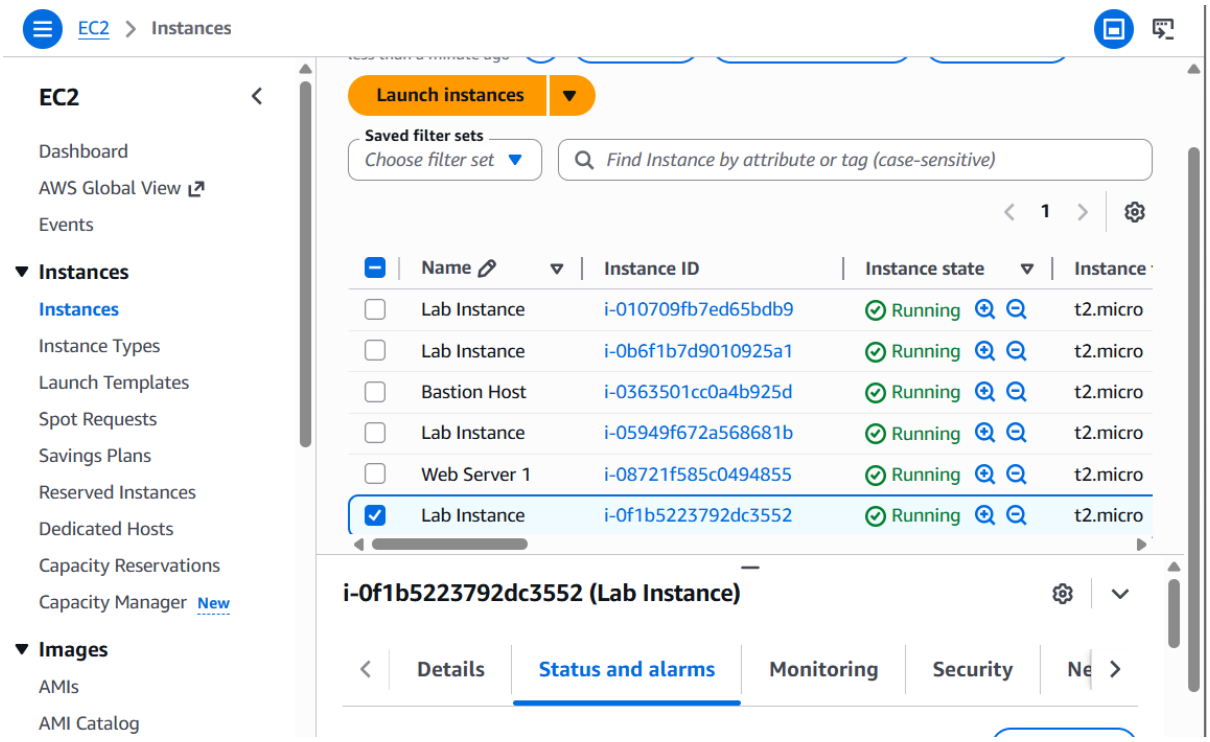




select EC2

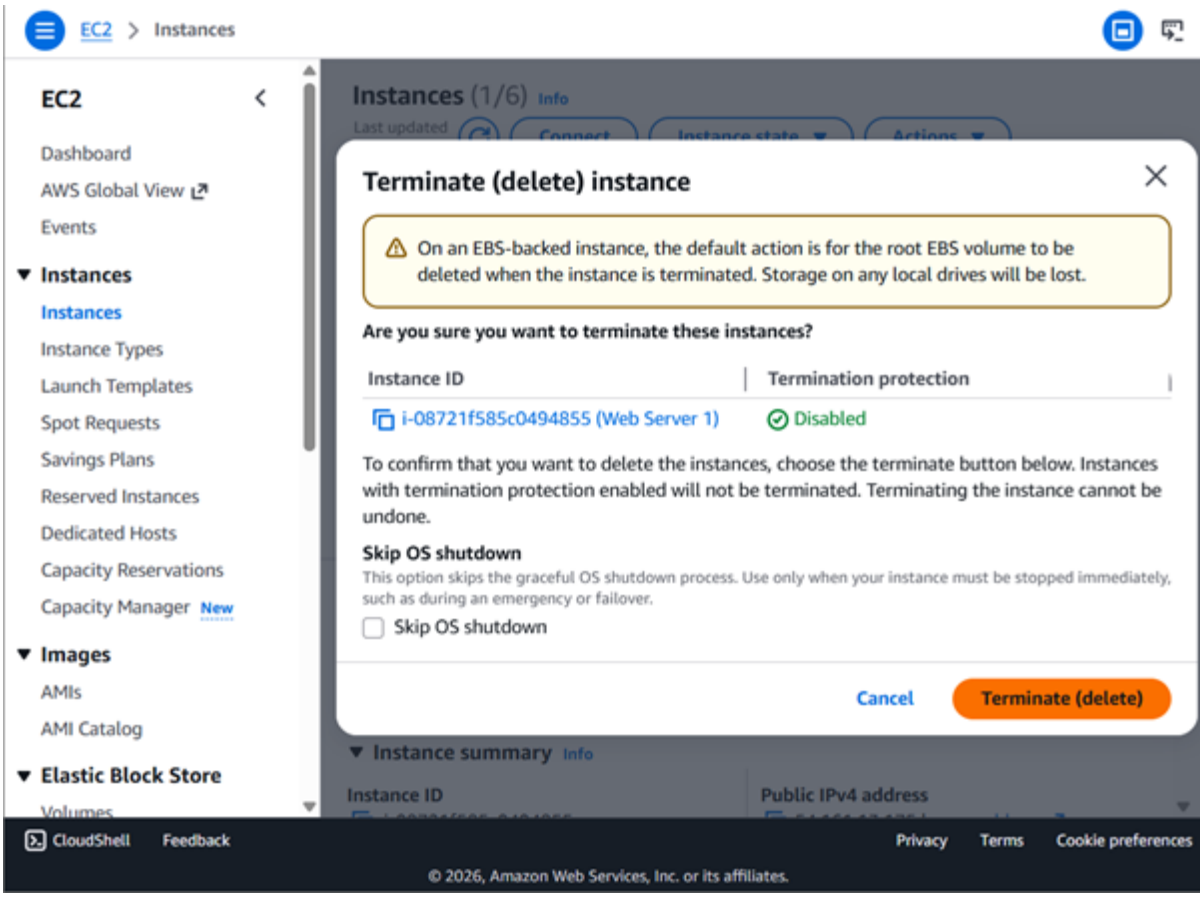
choose Instances

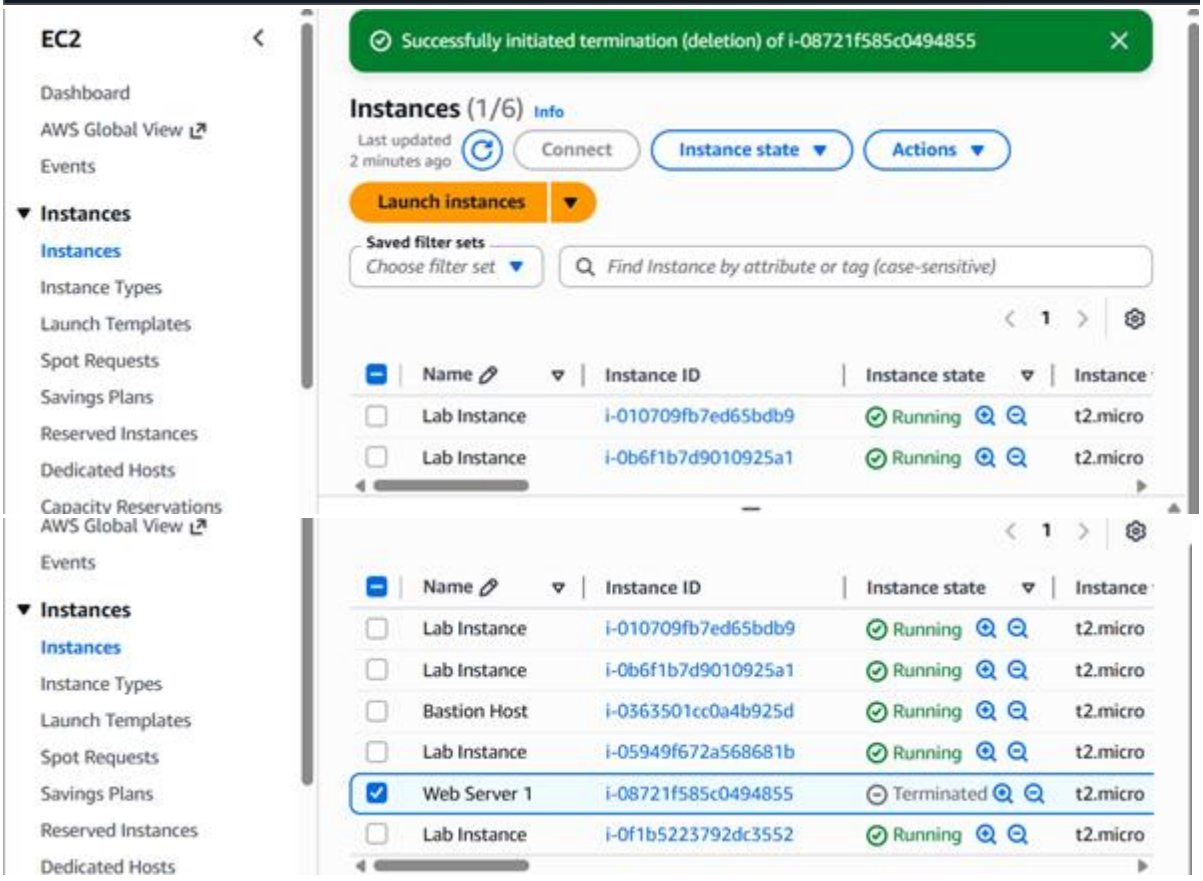




Terminate Web Server 1

choose Instance State > Terminate Instance


EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instances (1/6) Info
Terminate (delete) instance
On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.
Are you sure you want to terminate these instances?
Instance ID
Termination protection
i-08721f585c0494855 (Web Server 1)
Disabled
To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.
Skip OS shutdown
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.
Skip OS shutdown
Cancel
Terminate (delete)
Instance summary Info
Instance ID
Public IPv4 address
CloudShell
Feedback
Privacy
Terms
Cookie preferences
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EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Successfully initiated termination (deletion) of i-08721f585c0494855
Instances (1/6) Info
Last updated 2 minutes ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance
Lab Instance i-010709fb7ed65bdb9 Running t2.micro
Lab Instance i-0b6f1b7d9010925a1 Running t2.micro
Bastion Host i-0363501cc0a4b925d Running t2.micro
Lab Instance i-05949f672a568681b Running t2.micro
Web Server 1 i-08721f585c0494855 Terminated t2.micro
Lab Instance i-0f1b5223792dc3552 Running t2.micro

- Task 4 – Reflection questions

1. Why was termination protection enabled while launching the EC2 instance?

Answer: Termination protection was enabled to prevent the instance from being accidentally terminated either by a user mistake, an automated process, or an unintended API call. Since the EC2 instance serves as the foundation for the entire lab (hosting the web application, being used to create an AMI, and forming the base of the Auto Scaling setup), accidental termination would mean losing all configuration work done on it. With termination protection on, any termination attempt is blocked unless the protection is explicitly disabled first, adding a critical safety layer during the setup phase

2. Which metrics did you monitor to check the EC2 instance performance?

Answer: Through Amazon CloudWatch and the EC2 Monitoring tab, the key metrics monitored included:
- CPU Utilization: the percentage of allocated compute units in use, indicating whether the instance is under load or idle.
- Network In / Network Out: the volume of incoming and outgoing network traffic, useful for detecting traffic spikes or connectivity issues.
- Disk Read / Disk Write (I/O) : the read and write operations on the instance's storage, helping identify I/O bottlenecks.
- Status Check metrics: both system status checks (AWS infrastructure health) and instance status checks (OS-level health), confirming the instance is reachable and operational.

3. Why did you modify the security group to allow HTTP access?

Answer: By default, a new security group only allows outbound traffic and blocks all inbound traffic. HTTP access (port 80) had to be explicitly allowed so that users and the Load Balancer could reach the web server running on the instance through a browser. Without this inbound rule, all web requests would be silently dropped by the firewall, and the application would be unreachable even if the web server was running perfectly inside the instance. Security groups act as virtual firewalls, so every required port must be opened intentionally

4. What is the purpose of using a Load Balancer in this lab?

Answer: The Load Balancer serves several purposes:

- Traffic distribution evenly spreads incoming HTTP requests across all healthy EC2 instances in the Auto Scaling group, preventing any single instance from being overwhelmed.
- High availability routes traffic only to healthy instances (health checks), so if one instance fails, users are transparently redirected to a working one.
- Single entry point provides one stable DNS endpoint for the application, even as instances behind it are added or removed dynamically by Auto Scaling.
- Decoupling: users never interact with individual instances directly, making the architecture more scalable and fault-tolerant.

5. What happens in the Auto Scaling group when application traffic increases?

Answer: When traffic increases and a defined scaling metric threshold is crossed, the Auto Scaling group triggers a scale out action:
- It launches new EC2 instances using the configured Launch Template. The new instances are automatically registered with the Load Balancer.
- Traffic is redistributed across all healthy instances, reducing the load on each one.

Once traffic subsides and metrics drop below the threshold, the group performs a scale-in terminating excess instances down to the minimum count optimizing cost. At all times, the group ensures the number of running instances stays within the defined minimum and maximum boundaries

6. Why did you create an Amazon Machine Image (AMI) before creating the launch template?

Answer: The AMI was created first because it serves as the blueprint that the Launch Template references. An AMI is a snapshot of the EC2 instance at a specific point in time capturing the operating system, installed software in a ready to run state.
When Auto Scaling needs to launch a new instance during a scale out event, it uses the Launch Template, which points to that AMI. This guarantees that every new instance launched is identical to the original pre configured, pre installed, and immediately ready to serve traffic without any manual setup. Without the AMI, the Launch Template would have no consistent image to base new instances on, making automated and repeatable scaling impossible

1. Task Completion (70 points total)

Each AWS task is evaluated based on correctness, completeness, and evidence (e.g., screenshots output).

- Task 1 – Launch EC2 via Console
- Task 2 – Monitor and Manage activity such as status checks, troubleshoot, system log, CPU Utilization, Network In, Network Out, trigger the alarm, manage security group, resize the instance, stop and terminate the instance.
- Task 3 – Load Balancer & Auto Scaling Group. Such as Create an AMI for Auto Scaling, create a load balancer, create a launch template and an auto scaling group, verify that load balancing is working, test auto scaling, Terminate Web Server 1.

- Task 4 – Reflection Questions. Complete and thoughtful answers for all questions
    1. Why was termination protection enabled while launching the EC2 instance?
    2. Which metrics did you monitor to check the EC2 instance performance?
    3. Why did you modify the security group to allow HTTP access?
    4. What is the purpose of using a Load Balancer in this lab?
    5. What happens in the Auto Scaling group when application traffic increases?
    6. Why did you create an Amazon Machine Image (AMI) before creating the launch template?

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